JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Definite Integration

JEE Main Mathematics · Previous Year Questions with Answer Key

331	180	2002 – 2026	22	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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Numerical	77 questions
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Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

MCQ (Single Correct Answer) 231 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

The value of the integral $\int_0^2 \frac{\sqrt{x(x^2+x+1)}}{(\sqrt{x+1})\left(\sqrt{x^4+x^2+1}\right)}\mathrm{d}x$ is equal to:

(A) $\frac{1}{3}\log_e(3-2\sqrt{2})$ (B) $\frac{2}{3}\log_e(4+\sqrt{2})$

(C) $\frac{2}{3}\log_e(3+2\sqrt{2})$ (D) $\frac{1}{3}\log_e(1+6\sqrt{2})$

Q2. JEE Main 2026 (Online) 8th April Evening Shift

Let $f(x)=\begin{cases}\frac{1}{3}, & x\le \pi/2\\ \frac{\mathrm{b}(1-\sin x)}{(\pi-2x)^2}, & x>\pi/2\end{cases}$. If f is continuous at $x=\pi/2$, then the value of $\int_0^{3\,\mathrm{b}-6}|x^2+2x-3|\mathrm{d}x$ is :

(A) 5 (B) 2 (C) 3 (D) 4

Q3. JEE Main 2026 (Online) 6th April Evening Shift

Let $\mathrm{A}=\begin{bmatrix}1 & 3 & -1\\ 2 & 1 & \alpha\\ 0 & 1 & -1\end{bmatrix}$ be a singular matrix. Let $f(x)=\int_0^x(\mathrm{t}^2+2\mathrm{t}+3)\mathrm{dt}, x\in[1,\alpha]$. If M and m are respectively the maximum and the minimum values of f in $[1,\alpha]$, then $3(\mathrm{M}-\mathrm{m})$ is equal to :

(A) 64 (B) 68 (C) 72 (D) 76

Q4. JEE Main 2026 (Online) 6th April Evening Shift

The value of the integral $\int_{-1}^{1}\left(\frac{x^3+|x|+1}{x^2+2|x|+1}\right)\mathrm{d}x$ is equal to:

(A) $3\log_e 2$ (B) $2\log_e 2$ (C) $5\log_e 3$ (D) $3\log_e 3$

Q5. JEE Main 2026 (Online) 6th April Morning Shift

The value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\frac{32\cos^4 x}{1+e^{\sin x}}\right)dx$ is :

(A) $4\pi+2$ (B) $3\pi+8$ (C) $3\pi+4$ (D) $4\pi+3$

Q6. JEE Main 2026 (Online) 5th April Evening Shift

Let $\left(2^{1-\mathrm{a}}+2^{1+\mathrm{a}}\right), f(\mathrm{a}), (3^{\mathrm{a}}+3^{-\mathrm{a}})$ be in A.P. and α be the minimum value of $f(\mathrm{a})$. Then the value of the integral $\int_{\log_e(\alpha-1)}^{\log_e(\alpha)}\frac{dx}{(e^{2x}-e^{-2x})}$ is :

(A) $\frac{1}{2}\log_e\left(\frac{4}{3}\right)$ (B) $\frac{1}{4}\log_e\left(\frac{4}{3}\right)$

(C) $\frac{1}{2}\log_e\left(\frac{8}{5}\right)$ (D) $\frac{1}{4}\log_e\left(\frac{8}{5}\right)$

Q7. JEE Main 2026 (Online) 5th April Morning Shift

The value of the integral $\int_0^{\infty} \frac{\log_e(x)}{x^2+4}dx$ is:

(A) $\frac{\pi \log_e(2)}{2}$ (B) $\frac{\pi \log_e(2)}{4}$ (C) $1+\pi \log_e(2)$ (D) $2+\pi \log_e(2)$

Q8. JEE Main 2026 (Online) 5th April Morning Shift

The value of the integral $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \left(\frac{4-\text{cosec}^2 x}{\cos^4 x}\right)dx$ is :

(A) $\frac{11}{\sqrt{3}}$ (B) $\frac{16}{\sqrt{3}}$ (C) $\frac{32}{3\sqrt{3}}$ (D) $\frac{64}{3\sqrt{3}}$

Q9. JEE Main 2026 (Online) 4th April Evening Shift

If $\alpha = 1$ and $\beta = 1+i\sqrt{2}$, where $i = \sqrt{-1}$ are two roots of the equation $x^3+ax^2+bx+c=0, a, b, \in \mathbb{R}$, then $\int_{-1}^{1}(x^3+ax^2+bx+c)dx$ is equal to:

(A) -2 (B) -4 (C) -8 (D) -10

Q10. JEE Main 2026 (Online) 4th April Evening Shift

The integral $\int_0^1 \cot^{-1}\left(1+x+x^2\right)dx$ is equal to :

(A) $2\tan^{-1}2+\frac{1}{2}\log_e\left(\frac{5}{4}\right)+\frac{\pi}{2}$ (B) $2\tan^{-1}2+\frac{1}{2}\log_e\left(\frac{5}{4}\right)-\frac{\pi}{2}$

(C) $2\tan^{-1}2-\frac{1}{2}\log_e\left(\frac{5}{4}\right)+\frac{\pi}{2}$ (D) $2\tan^{-1}2-\frac{1}{2}\log_e\left(\frac{5}{4}\right)-\frac{\pi}{2}$

Q11. JEE Main 2026 (Online) 4th April Morning Shift

Let f be a real polynomial of degree n such that $f(x) = f'(x)f''(x)$, for all $x \in \mathbb{R}$. If $f(0)=0$, then $36\left(f'(2)+f''(2)+\int_0^2 f(x)dx\right)$ is equal to:

(A) 42 (B) 46 (C) 56 (D) 66

Q12. JEE Main 2026 (Online) 4th April Morning Shift

Let $\int_{-2}^{2}(|\sin x|+[x\sin x])dx = 2(3-\cos 2)+\beta$, where $[\cdot]$ is the greatest integer function. Then $\beta \sin\left(\frac{\beta}{2}\right)$ equals:

(A) 1 (B) 2 (C) 4 (D) 8

Q13. JEE Main 2026 (Online) 2nd April Evening Shift

The value of $\int_0^{20\pi}(\sin^4 x+\cos^4 x)dx$ is equal to:

(A) $\frac{15\pi}{2}$ (B) 25π (C) 15π (D) $\frac{25\pi}{2}$

Q14. JEE Main 2026 (Online) 2nd April Morning Shift

Let [.] denote the greatest integer function. Then the value of $\int_0^3 \left(\frac{e^x+e^{-x}}{[x]!}\right)dx$ is :

(A) $e^2+e^3-\frac{1}{e^2}-\frac{1}{e^3}$

(B) $\frac{1}{2}\left(e^2+e^3-\frac{1}{e^2}-\frac{1}{e^3}\right)$

(C) $e^2+e^3-\frac{1}{2e^2}-\frac{1}{2e^3}$

(D) $\frac{1}{2}(e^2+e^3)-\frac{1}{e^2}-\frac{1}{e^3}$

Q15. JEE Main 2026 (Online) 28th January Evening Shift

Let [.] denote the greatest integer function. Then

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{12(3+[x])}{3+[\sin x]+[\cos x]}\right)dx$

is equal to :

(A) $12\pi+5$

(B) $11\pi+2$

(C) $15\pi+4$

(D) $13\pi+1$

Q16. JEE Main 2026 (Online) 28th January Morning Shift

Let f be a polynomial function such that $f\left(x^2+1\right)=x^4+5x^2+2$, for all $x\in\mathbb{R}$.

Then $\int_0^3 f(x)dx$ is equal to

(A) $\frac{33}{2}$

(B) $\frac{5}{3}$

(C) $\frac{27}{2}$

(D) $\frac{41}{3}$

Q17. JEE Main 2026 (Online) 23rd January Morning Shift

The value of the integral $\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1+\sqrt[3]{\tan 2x}}$ is :

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{18}$

(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{12}$

Q18. JEE Main 2026 (Online) 22nd January Morning Shift

The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1}{[x]+4}\right)dx$, where $[\cdot]$ denotes the greatest integer function, is

(A) $\frac{1}{60}(21\pi-1)$

(B) $\frac{1}{60}(\pi-7)$

(C) $\frac{7}{60}(\pi-3)$

(D) $\frac{7}{60}(3\pi-1)$

Q19. JEE Main 2026 (Online) 22nd January Morning Shift

Let $f:[1,\infty)\to\mathbb{R}$ be a differentiable function. If $6\int_1^x f(t)dt=3xf(x)+x^3-4$ for all $x\geq 1$, then the value of $f(2)-f(3)$ is :

(A) 4

(B) 3

(C) -4

(D) -3

Q20. JEE Main 2026 (Online) 21st January Morning Shift

The value of $\int_{-\pi/6}^{\pi/6} \left(\frac{\pi+4x^{11}}{1-\sin(|x|+\pi/6)}\right)dx$ is equal to:

(A) 8π (B) 4π (C) 2π (D) 6π

Q21. JEE Main 2025 (Online) 8th April Evening Shift

The integral $\int_{-1}^{\frac{3}{2}} \left(|\pi^2 x \sin(\pi x)|\right)dx$ is equal to:

(A) $2+3\pi$ (B) $4+\pi$ (C) $1+3\pi$ (D) $3+2\pi$

Q22. JEE Main 2025 (Online) 8th April Evening Shift

Let *f(x)* be a positive function and $I_1 = \int_{-\frac{1}{2}}^{1} 2x\, f(2x(1-2x))\, dx$ and $I_2 = \int_{-1}^{2} f(x(1-x))\, dx$. Then the value of $\frac{I_2}{I_1}$ is equal to _____

(A) 12 (B) 9 (C) 6 (D) 4

Q23. JEE Main 2025 (Online) 7th April Morning Shift

The integral $\int_0^{\pi} \frac{(x+3)\sin x}{1+3\cos^2 x}dx$ is equal to

(A) $\frac{\pi}{\sqrt{3}}(\pi+1)$ (B) $\frac{\pi}{3\sqrt{3}}(\pi+6)$ (C) $\frac{\pi}{\sqrt{3}}(\pi+2)$ (D) $\frac{\pi}{2\sqrt{3}}(\pi+4)$

Q24. JEE Main 2025 (Online) 4th April Evening Shift

Let $f(x)+2f\left(\frac{1}{x}\right)=x^2+5$ and $2g(x)-3g\left(\frac{1}{2}\right)=x, x>0$. If $\alpha=\int_1^2 f(x)\mathrm{d}x$, and $\beta=\int_1^2 g(x)\mathrm{d}x$, then the value of $9\alpha+\beta$ is :

(A) 0 (B) 10 (C) 1 (D) 11

Q25. JEE Main 2025 (Online) 4th April Morning Shift

The value of $\int_{-1}^{1} \frac{(1+\sqrt{|x|-x})e^x+(\sqrt{|x|-x})e^{-x}}{e^x+e^{-x}}dx$ is equal to

(A) $1+\frac{2\sqrt{2}}{3}$ (B) $1-\frac{2\sqrt{2}}{3}$ (C) $2+\frac{2\sqrt{2}}{3}$ (D) $3-\frac{2\sqrt{2}}{3}$

Q26. JEE Main 2025 (Online) 3rd April Evening Shift

The integral $\int_0^{\pi} \frac{8xdx}{4\cos^2 x+\sin^2 x}$ is equal to

(A) $2\pi^2$ (B) $4\pi^2$ (C) π^2 (D) $\frac{3\pi^2}{2}$

Q27. JEE Main 2025 (Online) 3rd April Morning Shift

Let the domain of the function $f(x)=\log_2\log_4\log_6\left(3+4x-x^2\right)$ be (a,b). If $\int_0^{b-a}\left[x^2\right]dx=p-\sqrt{q}-\sqrt{r}, p,q,r\in\mathbb{N}, \gcd(p,q,r)=1$, where $[\cdot]$ is the greatest integer function, then $p+q+r$ is equal to

(A) 10 (B) 11 (C) 9 (D) 8

Q28. JEE Main 2025 (Online) 2nd April Evening Shift

Let $f:[1,\infty)\to[2,\infty)$ be a differentiable function. If $10\int_1^x f(\mathrm{t})\mathrm{dt}=5xf(x)-x^5-9$ for all $x\geqslant 1$, then the value of $f(3)$ is :

(A) 22 (B) 26 (C) 32 (D) 18

Q29. JEE Main 2025 (Online) 2nd April Evening Shift

Let (a,b) be the point of intersection of the curve $x^2=2y$ and the straight line $y-2x-6=0$ in the second quadrant. Then the integral $\mathrm{I}=\int_{\mathrm{a}}^{\mathrm{b}}\frac{9x^2}{1+5^x}\,\mathrm{d}x$ is equal to :

(A) 27 (B) 18 (C) 24 (D) 21

Q30. JEE Main 2025 (Online) 2nd April Evening Shift

$4\int_0^1\left(\frac{1}{\sqrt{3+x^2}+\sqrt{1+x^2}}\right)dx-3\log_e(\sqrt{3})$ is equal to :

(A) $2-\sqrt{2}-\log_{\mathrm{e}}(1+\sqrt{2})$ (B) $2+\sqrt{2}+\log_{\mathrm{e}}(1+\sqrt{2})$

(C) $2+\sqrt{2}-\log_{\mathrm{e}}(1+\sqrt{2})$ (D) $2-\sqrt{2}+\log_e(1+\sqrt{2})$

Q31. JEE Main 2025 (Online) 29th January Evening Shift

Let $f(x)=\int\limits_0^x \mathrm{t}\left(\mathrm{t}^2-9\mathrm{t}+20\right)\mathrm{dt}, 1\leq x\leq 5$. If the range of f is $[\alpha,\beta]$, then $4(\alpha+\beta)$ equals :

(A) 253 (B) 157 (C) 154 (D) 125

Q32. JEE Main 2025 (Online) 29th January Morning Shift

The integral $80\int\limits_0^{\frac{\pi}{4}}\left(\frac{\sin\theta+\cos\theta}{9+16\sin 2\theta}\right)d\theta$ is equal to :

(A) $3\log 4$ (B) $4\log 3$ (C) $6\log\frac{4}{3}$ (D) $2\log 3$

Q33. JEE Main 2025 (Online) 28th January Evening Shift

Let $\mathrm{f}:\mathrm{R}\to\mathrm{R}$ be a twice differentiable function such that $f(2)=1$. If $\mathrm{F(x)}=\mathrm{x}f(\mathrm{x})$ for all $\mathrm{x}\in\mathrm{R}$, $\int\limits_0^2 xF'(x)dx=6$ and $\int\limits_0^2 x^2F''(x)dx=40$, then $F'(2)+\int\limits_0^2 F(x)dx$ is equal to :

(A) 13 (B) 11 (C) 9 (D) 15

Q34. JEE Main 2025 (Online) 28th January Evening Shift

Let f be a real valued continuous function defined on the positive real axis such that $g(x)=\int_0^x tf(t)dt$. If $g\left(x^3\right)=x^6+x^7$, then value of $\sum_{r=1}^{15} f\left(r^3\right)$ is :

(A) 270 (B) 340 (C) 310 (D) 320

Q35. JEE Main 2025 (Online) 28th January Morning Shift

If $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{96x^2\cos^2 x}{(1+e^x)}\mathrm{d}x=\pi\left(\alpha\pi^2+\beta\right), \alpha,\beta\in\mathbb{Z}$, then $(\alpha+\beta)^2$ equals

(A) 196 (B) 100 (C) 64 (D) 144

Q36. JEE Main 2025 (Online) 24th January Morning Shift

If $I(m,n)=\int_0^1 x^{m-1}(1-x)^{n-1}dx, m,n>0$, then $I(9,14)+I(10,13)$ is

(A) $I(9,1)$ (B) $I(1,13)$ (C) $\mathrm{I}(19,27)$ (D) $\mathrm{I}(9,13)$

Q37. JEE Main 2025 (Online) 23rd January Evening Shift

If $\mathrm{I}=\int_0^{\frac{\pi}{2}} \frac{\sin^{\frac{3}{2}}x}{\sin^{\frac{3}{2}}x+\cos^{\frac{3}{2}}x}\,\mathrm{d}x$, then $\int_0^{2\mathrm{I}} \frac{x\sin x\cos x}{\sin^4 x+\cos^4 x}\,\mathrm{d}x$ equals :

(A) $\frac{\pi^2}{12}$ (B) $\frac{\pi^2}{4}$ (C) $\frac{\pi^2}{16}$ (D) $\frac{\pi^2}{8}$

Q38. JEE Main 2025 (Online) 23rd January Morning Shift

The value of $\int_{e^2}^{e^4} \frac{1}{x}\left(\frac{e^{\left((\log_e x)^2+1\right)^{-1}}}{e^{\left((\log_e x)^2+1\right)^{-1}}+e^{\left((6-\log_e x)^2+1\right)^{-1}}}\right)dx$ is

(A) 1 (B) $\log_e 2$ (C) e^2 (D) 2

Q39. JEE Main 2025 (Online) 22nd January Morning Shift

Let for $f(x)=7\tan^8 x+7\tan^6 x-3\tan^4 x-3\tan^2 x$, $\mathrm{I}_1=\int_0^{\pi/4} f(x)\mathrm{d}x$ and $\mathrm{I}_2=\int_0^{\pi/4} xf(x)\mathrm{d}x$. Then $7\mathrm{I}_1+12\mathrm{I}_2$ is equal to :

(A) 2π (B) 1 (C) π (D) 2

Q40. JEE Main 2024 (Online) 9th April Evening Shift

The integral $\int_{1/4}^{3/4} \cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$ is equal to

(A) $-1/2$ (B) $-1/4$ (C) 1/4 (D) 1/2

Q41. JEE Main 2024 (Online) 9th April Evening Shift

$\lim\limits_{x\to\frac{\pi}{2}}\left(\frac{\int_{x^3}^{(\pi/2)^3}\left(\sin\left(2t^{1/3}\right)+\cos\left(t^{1/3}\right)\right)dt}{\left(x-\frac{\pi}{2}\right)^2}\right)$ is equal to

(A) $\frac{3\pi^2}{2}$ (B) $\frac{9\pi^2}{8}$ (C) $\frac{5\pi^2}{9}$ (D) $\frac{11\pi^2}{10}$

Q42. JEE Main 2024 (Online) 9th April Evening Shift

The value of the integral $\int_{-1}^{2}\log_e\left(x+\sqrt{x^2+1}\right)dx$ is

(A) $\sqrt{5}-\sqrt{2}+\log_e\left(\frac{7+4\sqrt{5}}{1+\sqrt{2}}\right)$ (B) $\sqrt{2}-\sqrt{5}+\log_e\left(\frac{7+4\sqrt{5}}{1+\sqrt{2}}\right)$

(C) $\sqrt{5}-\sqrt{2}+\log_e\left(\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right)$ (D) $\sqrt{2}-\sqrt{5}+\log_e\left(\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right)$

Q43. JEE Main 2024 (Online) 8th April Evening Shift

Let $\int_{\alpha}^{\log_e 4}\frac{dx}{\sqrt{e^x-1}}=\frac{\pi}{6}$. Then e^{α} and $e^{-\alpha}$ are the roots of the equation :

(A) $2x^2-5x+2=0$ (B) $x^2-2x-8=0$ (C) $2x^2-5x-2=0$ (D) $x^2+2x-8=0$

Q44. JEE Main 2024 (Online) 8th April Morning Shift

The value of $k\in\mathbb{N}$ for which the integral $I_n=\int_0^1\left(1-x^k\right)^n dx, n\in\mathbb{N}$, satisfies $147I_{20}=148I_{21}$ is

(A) 8 (B) 14 (C) 7 (D) 10

Q45. JEE Main 2024 (Online) 6th April Morning Shift

$\int_0^{\pi/4}\frac{\cos^2 x\sin^2 x}{\left(\cos^3 x+\sin^3 x\right)^2}dx$ is equal to

(A) 1/9 (B) 1/6 (C) 1/3 (D) 1/12

Q46. JEE Main 2024 (Online) 5th April Evening Shift

Let $\beta(\mathrm{m},\mathrm{n})=\int_0^1 x^{\mathrm{m}-1}(1-x)^{\mathrm{n}-1}\,dx,\ \mathrm{m},\mathrm{n}>0$. If $\int_0^1\left(1-x^{10}\right)^{20}dx=\mathrm{a}\times\beta(\mathrm{b},\mathrm{c})$, then $100(\mathrm{a}+\mathrm{b}+\mathrm{c})$ equals _____.

(A) 2012 (B) 1021 (C) 1120 (D) 2120

Q47. JEE Main 2024 (Online) 5th April Morning Shift

The integral $\int_0^{\pi/4}\frac{136\sin x}{3\sin x+5\cos x}\,dx$ is equal to :

(A) $3\pi-50\log_e 2+20\log_e 5$ (B) $3\pi-25\log_e 2+10\log_e 5$

(C) $3\pi-10\log_e(2\sqrt{2})+10\log_e 5$ (D) $3\pi-30\log_e 2+20\log_e 5$

Q48. JEE Main 2024 (Online) 5th April Morning Shift

The value of $\int_{-\pi}^{\pi} \frac{2y(1+\sin y)}{1+\cos^2 y} dy$ is :

(A) $\frac{\pi}{2}$ (B) π^2 (C) $\frac{\pi^2}{2}$ (D) $2\pi^2$

Q49. JEE Main 2024 (Online) 4th April Evening Shift

Let $f(x) = \int_0^x \left(t + \sin\left(1 - e^t\right)\right) dt, x \in \mathbb{R}$. Then, $\lim_{x \to 0} \frac{f(x)}{x^3}$ is equal to

(A) $\frac{1}{6}$ (B) $-\frac{1}{6}$ (C) $\frac{2}{3}$ (D) $-\frac{2}{3}$

Q50. JEE Main 2024 (Online) 4th April Evening Shift

If the value of the integral $\int_{-1}^{1} \frac{\cos \alpha x}{1+3^x} dx$ is $\frac{2}{\pi}$, then a value of α is

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Q51. JEE Main 2024 (Online) 4th April Morning Shift

Let $f(x) = \begin{cases} -2, & -2 \leq x \leq 0 \\ x - 2, & 0 < x \leq 2 \end{cases}$ and $\mathrm{h}(x) = f(|x|) + |f(x)|$. Then $\int_{-2}^{2} \mathrm{h}(x) \mathrm{d}x$ is equal to:

(A) 2 (B) 6 (C) 4 (D) 1

Q52. JEE Main 2024 (Online) 1st February Evening Shift

If $\int_0^{\frac{\pi}{3}} \cos^4 x \, \mathrm{d}x = \mathrm{a}\pi + \mathrm{b}\sqrt{3}$, where a and b are rational numbers, then $9\mathrm{a} + 8\mathrm{b}$ is equal to :

(A) 2 (B) 1 (C) 3 (D) $\frac{3}{2}$

Q53. JEE Main 2024 (Online) 1st February Evening Shift

The value of $\int_0^1 \left(2x^3 - 3x^2 - x + 1\right)^{\frac{1}{3}} \mathrm{d}x$ is equal to :

(A) -1 (B) 2 (C) 0 (D) 1

Q54. JEE Main 2024 (Online) 1st February Morning Shift

The value of the integral $\int_0^{\pi/4} \frac{x \, \mathrm{d}x}{\sin^4(2x) + \cos^4(2x)}$ equals :

(A) $\frac{\sqrt{2}\pi^2}{8}$ (B) $\frac{\sqrt{2}\pi^2}{16}$ (C) $\frac{\sqrt{2}\pi^2}{32}$ (D) $\frac{\sqrt{2}\pi^2}{64}$

Q55. JEE Main 2024 (Online) 31st January Evening Shift

Let $f, g : (0, \infty) \to \mathbb{R}$ be two functions defined by $f(x) = \int_{-x}^{x} \left(|t| - t^2\right) e^{-t^2} dt$ and $g(x) = \int_{0}^{x^2} t^{1/2} e^{-t} dt$. Then, the value of $9\left(f\left(\sqrt{\log_e 9}\right) + g\left(\sqrt{\log_e 9}\right)\right)$ is equal to :

(A) 10 (B) 9 (C) 8 (D) 6

Q56. JEE Main 2024 (Online) 30th January Evening Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \frac{x}{(1+x^4)^{1/4}}$, and $g(x) = f(f(f(f(x))))$. Then, $18 \int_0^{\sqrt{2\sqrt{5}}} x^2 g(x) dx$ is equal to

(A) 36 (B) 33 (C) 39 (D) 42

Q57. JEE Main 2024 (Online) 30th January Evening Shift

Let $y = f(x)$ be a thrice differentiable function in $(-5, 5)$. Let the tangents to the curve $y = f(x)$ at $(1, f(1))$ and $(3, f(3))$ make angles $\pi/6$ and $\pi/4$, respectively with positive x-axis. If $27 \int_1^3 \left((f'(t))^2 + 1\right) f''(t) dt = \alpha + \beta\sqrt{3}$ where α, β are integers, then the value of $\alpha + \beta$ equals

(A) 26 (B) -16 (C) 36 (D) -14

Q58. JEE Main 2024 (Online) 30th January Evening Shift

Let a and b be real constants such that the function f defined by $f(x) = \begin{cases} x^2 + 3x + a & , x \le 1 \\ bx + 2 & , x > 1 \end{cases}$ be differentiable on $\mathbb{R}$. Then, the value of $\int_{-2}^{2} f(x) dx$ equals

(A) 21 (B) 19/6 (C) 17 (D) 15/6

Q59. JEE Main 2024 (Online) 30th January Evening Shift

Let $\mathrm{f} : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = ae^{2x} + be^x + cx$. If $f(0) = -1$, $f'(\log_e 2) = 21$ and $\int_0^{\log_e 4} (f(x) - cx) dx = \frac{39}{2}$, then the value of $|a + b + c|$ equals

(A) 16 (B) 12 (C) 8 (D) 10

Q60. JEE Main 2024 (Online) 30th January Morning Shift

The value of $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{n^3}{(n^2+k^2)(n^2+3k^2)}$ is :

(A) $\frac{\pi}{8(2\sqrt{3}+3)}$ (B) $\frac{(2\sqrt{3}+3)\pi}{24}$ (C) $\frac{13\pi}{8(4\sqrt{3}+3)}$ (D) $\frac{13(2\sqrt{3}-3)\pi}{8}$

Q61. JEE Main 2024 (Online) 30th January Morning Shift

Let $f:\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow \mathbf{R}$ be a differentiable function such that $f(0)=\frac{1}{2}$. If the $\lim\limits_{x \rightarrow 0} \frac{x \int_0^x f(t) dt}{e^{x^2}-1}=\alpha$, then $8 \alpha^2$ is equal to :

(A) 4 (B) 2 (C) 1 (D) 16

Q62. JEE Main 2024 (Online) 29th January Morning Shift

$\lim\limits_{x \rightarrow \frac{\pi}{2}}\left(\frac{1}{\left(x-\frac{\pi}{2}\right)^2} \int\limits_{x^3}^{\left(\frac{\pi}{2}\right)^3} \cos \left(t^{\frac{1}{3}}\right) dt\right)$ is equal to

(A) $\frac{3\pi^2}{4}$ (B) $\frac{3\pi^2}{8}$ (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{8}$

Q63. JEE Main 2024 (Online) 29th January Morning Shift

If the value of the integral $\int\limits_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{x^2 \cos x}{1+\pi^x}+\frac{1+\sin^2 x}{1+e^{\sin x^{2023}}}\right) dx=\frac{\pi}{4}(\pi+a)-2$, then the value of a is

(A) $-\frac{3}{2}$ (B) 3 (C) $\frac{3}{2}$ (D) 2

Q64. JEE Main 2024 (Online) 27th January Evening Shift

For $0<\mathrm{a}<1$, the value of the integral $\int\limits_0^{\pi} \frac{\mathrm{d}x}{1-2\mathrm{a} \cos x+\mathrm{a}^2}$ is :

(A) $\frac{\pi^2}{\pi+a^2}$ (B) $\frac{\pi^2}{\pi-a^2}$ (C) $\frac{\pi}{1-\mathrm{a}^2}$ (D) $\frac{\pi}{1+\mathrm{a}^2}$

Q65. JEE Main 2024 (Online) 27th January Morning Shift

If $\int\limits_0^1 \frac{1}{\sqrt{3+x}+\sqrt{1+x}} \mathrm{d}x=\mathrm{a}+\mathrm{b} \sqrt{2}+\mathrm{c} \sqrt{3}$, where $\mathrm{a}, \mathrm{b}, \mathrm{c}$ are rational numbers, then $2\mathrm{a}+3\,\mathrm{b}-4\mathrm{c}$ is equal to :

(A) 10 (B) 7 (C) 4 (D) 8

Q66. JEE Main 2024 (Online) 27th January Morning Shift

If (a, b) be the orthocentre of the triangle whose vertices are $(1,2),(2,3)$ and $(3,1)$, and $\mathrm{I}_1=\int\limits_{\mathrm{a}}^{\mathrm{b}} x \sin \left(4x-x^2\right) \mathrm{d}x, \mathrm{I}_2=\int\limits_{\mathrm{a}}^{\mathrm{b}} \sin \left(4x-x^2\right) \mathrm{d}x$, then $36 \frac{\mathrm{I}_1}{\mathrm{I}_2}$ is equal to :

(A) 80 (B) 72 (C) 66 (D) 88

Q67. JEE Main 2023 (Online) 15th April Morning Shift

If $\int\limits_0^1 \frac{1}{(5+2x-2x^2)\left(1+e^{(2-4x)}\right)} dx=\frac{1}{\alpha} \log_e\left(\frac{\alpha+1}{\beta}\right), \alpha, \beta>0$, then $\alpha^4-\beta^4$ is equal to :

(A) -21 (B) 21 (C) 19 (D) 0

Q68. JEE Main 2023 (Online) 13th April Evening Shift

The value of $\frac{e^{-\frac{\pi}{4}}+\int_0^{\frac{\pi}{4}} e^{-x}\tan^{50}x\,dx}{\int_0^{\frac{\pi}{4}} e^{-x}(\tan^{49}x+\tan^{51}x)\,dx}$ is

(A) 51 (B) 50 (C) 25 (D) 49

Q69. JEE Main 2023 (Online) 13th April Morning Shift

$\int_0^{\infty} \frac{6}{e^{3x}+6e^{2x}+11e^{x}+6}dx =$

(A) $\log_e\left(\frac{256}{81}\right)$ (B) $\log_e\left(\frac{64}{27}\right)$

(C) $\log_e\left(\frac{32}{27}\right)$ (D) $\log_e\left(\frac{512}{81}\right)$

Q70. JEE Main 2023 (Online) 11th April Evening Shift

If $f:\mathbb{R}\to\mathbb{R}$ be a continuous function satisfying $\int_0^{\frac{\pi}{2}} f(\sin 2x)\sin x\,dx+\alpha\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x\,dx=0$, then the value of α is :

(A) $-\sqrt{3}$ (B) $\sqrt{2}$ (C) $-\sqrt{2}$ (D) $\sqrt{3}$

Q71. JEE Main 2023 (Online) 11th April Evening Shift

Let the function $f:[0,2]\to\mathbb{R}$ be defined as

$$f(x)=\begin{cases} e^{\min\{x^2,x-[x]\}}, & x\in[0,1) \\ e^{[x-\log_e x]}, & x\in[1,2] \end{cases}$$

where $[t]$ denotes the greatest integer less than or equal to t. Then the value of the integral $\int_0^2 xf(x)dx$ is :

(A) $2e-1$ (B) $2e-\frac{1}{2}$

(C) $1+\frac{3e}{2}$ (D) $(e-1)\left(e^2+\frac{1}{2}\right)$

Q72. JEE Main 2023 (Online) 11th April Morning Shift

The value of the integral $\int_{-\log_e 2}^{\log_e 2} e^x\left(\log_e\left(e^x+\sqrt{1+e^{2x}}\right)\right)dx$ is equal to :

(A) $\log_e\left(\frac{(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)+\frac{\sqrt{5}}{2}$

(B) $\log_e\left(\frac{\sqrt{2}(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)-\frac{\sqrt{5}}{2}$

(C) $\log_e\left(\frac{2(2+\sqrt{5})}{\sqrt{1+\sqrt{5}}}\right)-\frac{\sqrt{5}}{2}$

(D) $\log_e\left(\frac{\sqrt{2}(3-\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)+\frac{\sqrt{5}}{2}$

Q73. JEE Main 2023 (Online) 10th April Evening Shift

Let f be a continuous function satisfying $\int_0^{t^2}\left(f(x)+x^2\right)dx=\frac{4}{3}t^3, \forall t>0$. Then $f\left(\frac{\pi^2}{4}\right)$ is equal to :

(A) $-\pi\left(1+\frac{\pi^3}{16}\right)$ (B) $\pi\left(1-\frac{\pi^3}{16}\right)$

(C) $-\pi^2\left(1+\frac{\pi^2}{16}\right)$ (D) $\pi^2\left(1-\frac{\pi^2}{16}\right)$

Q74. JEE Main 2023 (Online) 6th April Evening Shift

Let $f(x)$ be a function satisfying $f(x)+f(\pi-x)=\pi^2, \forall x\in\mathbb{R}$. Then $\int_0^{\pi} f(x)\sin x dx$ is equal to :

(A) π^2 (B) $\frac{\pi^2}{2}$ (C) $2\pi^2$ (D) $\frac{\pi^2}{4}$

Q75. JEE Main 2023 (Online) 6th April Morning Shift

Let $5f(x)+4f\left(\frac{1}{x}\right)=\frac{1}{x}+3, x>0$. Then $18\int_1^2 f(x)dx$ is equal to :

(A) $10\log_e 2+6$ (B) $5\log_e 2-3$ (C) $10\log_e 2-6$ (D) $5\log_e 2+3$

Q76. JEE Main 2023 (Online) 1st February Evening Shift

The value of the integral

$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{x+\frac{\pi}{4}}{2-\cos 2x}dx$ is :

(A) $\frac{\pi^2}{6\sqrt{3}}$ (B) $\frac{\pi^2}{6}$ (C) $\frac{\pi^2}{3\sqrt{3}}$ (D) $\frac{\pi^2}{12\sqrt{3}}$

Q77. JEE Main 2023 (Online) 31st January Evening Shift

Let $\alpha>0$. If $\int_0^{\alpha}\frac{x}{\sqrt{x+\alpha}-\sqrt{x}}\,dx=\frac{16+20\sqrt{2}}{15}$, then α is equal to :

(A) 4 (B) 2 (C) $2\sqrt{2}$ (D) $\sqrt{2}$

Q78. JEE Main 2023 (Online) 31st January Evening Shift

If $\phi(x)=\frac{1}{\sqrt{x}}\int_{\frac{\pi}{4}}^{x}\left(4\sqrt{2}\sin t-3\phi'(t)\right)dt, x>0$,

then $\emptyset'\left(\frac{\pi}{4}\right)$ is equal to :

(A) $\frac{4}{6+\sqrt{\pi}}$ (B) $\frac{4}{6-\sqrt{\pi}}$ (C) $\frac{8}{\sqrt{\pi}}$ (D) $\frac{8}{6+\sqrt{\pi}}$

Q79. JEE Main 2023 (Online) 31st January Morning Shift

Let $\alpha \in (0,1)$ and $\beta = \log_e(1-\alpha)$. Let $P_n(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \ldots + \frac{x^n}{n}, x \in (0,1)$. Then the integral $\int_0^{\alpha} \frac{t^{50}}{1-t} dt$ is equal to

(A) $-(\beta + P_{50}(\alpha))$

(B) $\beta - P_{50}(\alpha)$

(C) $P_{50}(\alpha) - \beta$

(D) $\beta + P_{50}(\alpha)$

Q80. JEE Main 2023 (Online) 31st January Morning Shift

The value of $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{(2+3\sin x)}{\sin x(1+\cos x)} dx$ is equal to :

(A) $\frac{10}{3} - \sqrt{3} + \log_e \sqrt{3}$

(B) $\frac{7}{2} - \sqrt{3} - \log_e \sqrt{3}$

(C) $\frac{10}{3} - \sqrt{3} - \log_e \sqrt{3}$

(D) $-2 + 3\sqrt{3} + \log_e \sqrt{3}$

Q81. JEE Main 2023 (Online) 30th January Morning Shift

If [t] denotes the greatest integer $\leq$ t, then the value of $\frac{3(e-1)}{e} \int_1^2 x^2 e^{[x]+[x^3]} dx$ is :

(A) $e^8 - e$

(B) $e^7 - 1$

(C) $e^9 - e$

(D) $e^8 - 1$

Q82. JEE Main 2023 (Online) 29th January Evening Shift

The value of the integral $\int_1^2 \left(\frac{t^4+1}{t^6+1}\right) dt$ is

(A) $\tan^{-1}\frac{1}{2} - \frac{1}{3}\tan^{-1}8 + \frac{\pi}{3}$

(B) $\tan^{-1}2 - \frac{1}{3}\tan^{-1}8 + \frac{\pi}{3}$

(C) $\tan^{-1}2 + \frac{1}{3}\tan^{-1}8 - \frac{\pi}{3}$

(D) $\tan^{-1}\frac{1}{2} + \frac{1}{3}\tan^{-1}8 - \frac{\pi}{3}$

Q83. JEE Main 2023 (Online) 29th January Evening Shift

The value of the integral $\int_{1/2}^{2} \frac{\tan^{-1}x}{x} dx$ is equal to :

(A) $\frac{\pi}{2}\log_e 2$

(B) $\frac{\pi}{4}\log_e 2$

(C) $\frac{1}{2}\log_e 2$

(D) $\pi \log_e 2$

Q84. JEE Main 2023 (Online) 29th January Morning Shift

Let $f(x) = x + \frac{a}{\pi^2-4}\sin x + \frac{b}{\pi^2-4}\cos x, x \in R$ be a function which

satisfies $f(x) = x + \int_0^{\pi/2} \sin(x+y) f(y) dy$. then $(a+b)$ is equal to

(A) $-2\pi(\pi+2)$

(B) $-\pi(\pi-2)$

(C) $-\pi(\pi+2)$

(D) $-2\pi(\pi-2)$

Q85. JEE Main 2023 (Online) 29th January Morning Shift

Let $[x]$ denote the greatest integer $\leq x$. Consider the function $f(x) = \max\{x^2, 1+[x]\}$. Then the value of the integral $\int_0^2 f(x)dx$ is

(A) $\frac{5+4\sqrt{2}}{3}$ (B) $\frac{4+5\sqrt{2}}{3}$ (C) $\frac{8+4\sqrt{2}}{3}$ (D) $\frac{1+5\sqrt{2}}{3}$

Q86. JEE Main 2023 (Online) 25th January Evening Shift

The integral $16\int_1^2 \frac{dx}{x^3(x^2+2)^2}$ is equal to

(A) $\frac{11}{12} + \log_e 4$ (B) $\frac{11}{6} + \log_e 4$ (C) $\frac{11}{12} - \log_e 4$ (D) $\frac{11}{6} - \log_e 4$

Q87. JEE Main 2023 (Online) 25th January Morning Shift

The minimum value of the function $f(x) = \int_0^2 e^{|x-t|}dt$ is :

(A) 2 (B) $2(e-1)$ (C) $e(e-1)$ (D) $2e-1$

Q88. JEE Main 2023 (Online) 24th January Evening Shift

$\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-4x^2}}dx$ is equal to :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) 2π

Q89. JEE Main 2022 (Online) 29th July Evening Shift

If $[t]$ denotes the greatest integer $\leq t$, then the value of $\int_0^1 \left[2x - \left|3x^2 - 5x + 2\right| + 1\right]\mathrm{d}x$ is :

(A) $\frac{\sqrt{37}+\sqrt{13}-4}{6}$ (B) $\frac{\sqrt{37}-\sqrt{13}-4}{6}$ (C) $\frac{-\sqrt{37}-\sqrt{13}+4}{6}$ (D) $\frac{-\sqrt{37}+\sqrt{13}+4}{6}$

Q90. JEE Main 2022 (Online) 29th July Morning Shift

The integral $\int_0^{\frac{\pi}{2}} \frac{1}{3+2\sin x+\cos x}\,\mathrm{d}x$ is equal to :

(A) $\tan^{-1}(2)$ (B) $\tan^{-1}(2) - \frac{\pi}{4}$

(C) $\frac{1}{2}\tan^{-1}(2) - \frac{\pi}{8}$ (D) $\frac{1}{2}$

Q91. JEE Main 2022 (Online) 29th July Morning Shift

If $f(\alpha) = \int_1^{\alpha} \frac{\log_{10} \mathrm{t}}{1+\mathrm{t}}\mathrm{dt}, \alpha > 0$, then $f\left(\mathrm{e}^3\right) + f\left(\mathrm{e}^{-3}\right)$ is equal to :

(A) 9 (B) $\frac{9}{2}$ (C) $\frac{9}{\log_e(10)}$ (D) $\frac{9}{2\log_e(10)}$

Q92. JEE Main 2022 (Online) 28th July Evening Shift

Let $I_n(x) = \int_0^x \frac{1}{(t^2+5)^n} dt, n = 1, 2, 3, \ldots$. Then :

(A) $50I_6 - 9I_5 = xI_5'$

(B) $50I_6 - 11I_5 = xI_5'$

(C) $50I_6 - 9I_5 = I_5'$

(D) $50I_6 - 11I_5 = I_5'$

Q93. JEE Main 2022 (Online) 28th July Morning Shift

The minimum value of the twice differentiable function $f(x) = \int\limits_0^x \mathrm{e}^{x-\mathrm{t}} f'(\mathrm{t})\mathrm{dt} - \left(x^2 - x + 1\right)\mathrm{e}^x$, $x \in \mathbf{R}$, is :

(A) $-\frac{2}{\sqrt{\mathrm{e}}}$

(B) $-2\sqrt{\mathrm{e}}$

(C) $-\sqrt{\mathrm{e}}$

(D) $\frac{2}{\sqrt{\mathrm{e}}}$

Q94. JEE Main 2022 (Online) 27th July Evening Shift

Let $f(x) = 2 + |x| - |x-1| + |x+1|, x \in \mathbf{R}$.

Consider

$(\mathrm{S1}) : f'\left(-\frac{3}{2}\right) + f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) = 2$

$(\mathrm{S2}) : \int\limits_{-2}^{2} f(x)\mathrm{d}x = 12$

Then,

(A) both (S1) and (S2) are correct

(B) both (S1) and (S2) are wrong

(C) only (S1) is correct

(D) only (S2) is correct

Q95. JEE Main 2022 (Online) 27th July Evening Shift

$\int\limits_0^2 \left(\left|2x^2 - 3x\right| + \left[x - \frac{1}{2}\right]\right)\mathrm{d}x$, where [t] is the greatest integer function, is equal to :

(A) $\frac{7}{6}$

(B) $\frac{19}{12}$

(C) $\frac{31}{12}$

(D) $\frac{3}{2}$

Q96. JEE Main 2022 (Online) 27th July Morning Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined as

$f(x) = a\sin\left(\frac{\pi[x]}{2}\right) + [2 - x], a \in \mathbb{R}$ where $[t]$ is the greatest integer less than or equal to t. If $\lim\limits_{x \to -1} f(x)$ exists, then the value of $\int\limits_0^4 f(x)dx$ is equal to

(A) -1

(B) -2

(C) 1

(D) 2

Q97. JEE Main 2022 (Online) 27th July Morning Shift

Let $I = \int_{\pi/4}^{\pi/3} \left(\frac{8\sin x - \sin 2x}{x}\right)dx$. Then

(A) $\frac{\pi}{2} < I < \frac{3\pi}{4}$

(B) $\frac{\pi}{5} < I < \frac{5\pi}{12}$

(C) $\frac{5\pi}{12} < I < \frac{\sqrt{2}}{3}\pi$

(D) $\frac{3\pi}{4} < I < \pi$

Q98. JEE Main 2022 (Online) 27th July Morning Shift

Let a function $f : \mathbb{R} \to \mathbb{R}$ be defined as :

$$f(x) = \begin{cases} \int_0^x (5 - |t-3|)dt, & x > 4 \\ x^2 + bx & , x \leq 4 \end{cases}$$

where $b \in \mathbb{R}$. If f is continuous at $x = 4$, then which of the following statements is NOT true?

(A) f is not differentiable at $x = 4$

(B) $f'(3) + f'(5) = \frac{35}{4}$

(C) f is increasing in $\left(-\infty, \frac{1}{8}\right) \cup (8, \infty)$

(D) f has a local minima at $x = \frac{1}{8}$

Q99. JEE Main 2022 (Online) 26th July Evening Shift

$\int_0^{20\pi} (|\sin x| + |\cos x|)^2 dx$ is equal to

(A) $10(\pi + 4)$ (B) $10(\pi + 2)$ (C) $20(\pi - 2)$ (D) $20(\pi + 2)$

Q100. JEE Main 2022 (Online) 25th July Evening Shift

Let $[t]$ denote the greatest integer less than or equal to t. Then the value of the integral $\int_{-3}^{101} \left([\sin(\pi x)] + e^{[\cos(2\pi x)]}\right) dx$ is equal to

(A) $\frac{52(1-e)}{e}$ (B) $\frac{52}{e}$ (C) $\frac{52(2+e)}{e}$ (D) $\frac{104}{e}$

Q101. JEE Main 2022 (Online) 25th July Morning Shift

For any real number x, let $[x]$ denote the largest integer less than equal to x. Let f be a real valued function defined on the interval $[-10, 10]$ by $f(x) = \begin{cases} x - [x], & \text{if } [x] \text{ is odd} \\ 1 + [x] - x, & \text{if } [x] \text{ is even} \end{cases}$. Then the value of $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x \, dx$ is :

(A) 4 (B) 2 (C) 1 (D) 0

Q102. JEE Main 2022 (Online) 29th June Evening Shift

Let f be a real valued continuous function on [0, 1] and $f(x) = x + \int_0^1 (x - t) f(t) dt$.

Then, which of the following points (x, y) lies on the curve y = f(x) ?

(A) (2, 4) (B) (1, 2) (C) (4, 17) (D) (6, 8)

Q103. JEE Main 2022 (Online) 29th June Evening Shift

If $\int_0^2 \left(\sqrt{2x} - \sqrt{2x - x^2}\right) dx = \int_0^1 \left(1 - \sqrt{1 - y^2} - \frac{y^2}{2}\right) dy + \int_1^2 \left(2 - \frac{y^2}{2}\right) dy + I$, then I equals

(A) $\int_0^1 \left(1 + \sqrt{1 - y^2}\right) dy$

(B) $\int_0^1 \left(\frac{y^2}{2} - \sqrt{1 - y^2} + 1\right) dy$

(C) $\int_0^1 \left(1 - \sqrt{1 - y^2}\right) dy$

(D) $\int_0^1 \left(\frac{y^2}{2} + \sqrt{1 - y^2} + 1\right) dy$

Q104. JEE Main 2022 (Online) 29th June Morning Shift

Let $f : R \to R$ be a function defined by :

$$f(x) = \begin{cases} \max\{t^3 - 3t : t \le x\} & ; \quad x \le 2 \\ x^2 + 2x - 6 & ; \quad 2 < x < 3 \\ [x - 3] + 9 & ; \quad 3 \le x \le 5 \\ 2x + 1 & ; \quad x > 5 \end{cases}$$

where [t] is the greatest integer less than or equal to t. Let m be the number of points where f is not differentiable and $I = \int_{-2}^{2} f(x)\, dx$. Then the ordered pair (m, I) is equal to :

(A) $\left(3, \frac{27}{4}\right)$

(B) $\left(3, \frac{23}{4}\right)$

(C) $\left(4, \frac{27}{4}\right)$

(D) $\left(4, \frac{23}{4}\right)$

Q105. JEE Main 2022 (Online) 29th June Morning Shift

$\int_0^5 \cos\left(\pi\left(x - \left[\frac{x}{2}\right]\right)\right) dx$,

where [t] denotes greatest integer less than or equal to t, is equal to:

(A) -3

(B) -2

(C) 2

(D) 0

Q106. JEE Main 2022 (Online) 28th June Evening Shift

Let f : R $\to$ R be a differentiable function such that $f\left(\frac{\pi}{4}\right) = \sqrt{2}$, $f\left(\frac{\pi}{2}\right) = 0$ and $f'\left(\frac{\pi}{2}\right) = 1$ and

let $g(x) = \int_x^{\pi/4} \left(f'(t) \sec t + \tan t \sec t\, f(t)\right) dt$ for $x \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right)$. Then $\lim\limits_{x \to \left(\frac{\pi}{2}\right)^-} g(x)$ is equal to :

(A) 2

(B) 3

(C) 4

(D) -3

Q107. JEE Main 2022 (Online) 28th June Evening Shift

Let f : R $\to$ R be a continuous function satisfying f(x) + f(x + k) = n, for all x $\in$ R where k > 0 and n is a positive integer. If $I_1 = \int_0^{4nk} f(x)dx$ and $I_2 = \int_{-k}^{3k} f(x)dx$, then :

(A) $I_1 + 2I_2 = 4nk$

(B) $I_1 + 2I_2 = 2nk$

(C) $I_1 + nI_2 = 4n^2k$

(D) $I_1 + nI_2 = 6n^2k$

Q108. JEE Main 2022 (Online) 28th June Morning Shift

Let [t] denote the greatest integer less than or equal to t. Then, the value of the integral $\int_0^1 [-8x^2 + 6x - 1]dx$ is equal to :

(A) -1 (B) $\frac{-5}{4}$ (C) $\frac{\sqrt{17}-13}{8}$ (D) $\frac{\sqrt{17}-16}{8}$

Q109. JEE Main 2022 (Online) 27th June Evening Shift

If m and n respectively are the number of local maximum and local minimum points of the function $f(x) = \int_0^{x^2} \frac{t^2-5t+4}{2+e^t}dt$, then the ordered pair (m, n) is equal to

(A) (3, 2) (B) (2, 3) (C) (2, 2) (D) (3, 4)

Q110. JEE Main 2022 (Online) 27th June Evening Shift

Let f be a differentiable function in $\left(0, \frac{\pi}{2}\right)$. If $\int_{\cos x}^1 t^2 f(t)dt = \sin^3 x + \cos x$, then $\frac{1}{\sqrt{3}} f'\left(\frac{1}{\sqrt{3}}\right)$ is equal to

(A) $6 - 9\sqrt{2}$ (B) $6 - \frac{9}{\sqrt{2}}$ (C) $\frac{9}{2} - 6\sqrt{2}$ (D) $\frac{9}{\sqrt{2}} - 6$

Q111. JEE Main 2022 (Online) 27th June Evening Shift

The integral $\int_0^1 \frac{1}{7^{\left[\frac{1}{x}\right]}}dx$, where [.] denotes the greatest integer function, is equal to

(A) $1 + 6\log_e\left(\frac{6}{7}\right)$ (B) $1 - 6\log_e\left(\frac{6}{7}\right)$

(C) $\log_e\left(\frac{7}{6}\right)$ (D) $1 - 7\log_e\left(\frac{6}{7}\right)$

Q112. JEE Main 2022 (Online) 27th June Morning Shift

The value of the integral

$\int_{-2}^2 \frac{|x^3+x|}{(e^{x|x|}+1)}dx$ is equal to :

(A) $5e^2$ (B) $3e^{-2}$ (C) 4 (D) 6

Q113. JEE Main 2022 (Online) 25th June Evening Shift

If $b_n = \int_0^{\frac{\pi}{2}} \frac{\cos^2 nx}{\sin x}dx$, $n \in N$, then

(A) $b_3 - b_2$, $b_4 - b_3$, $b_5 - b_4$ are in A.P. with common difference -2

(B) $\frac{1}{b_3-b_2}$, $\frac{1}{b_4-b_3}$, $\frac{1}{b_5-b_4}$ are in an A.P. with common difference 2

(C) $b_3 - b_2$, $b_4 - b_3$, $b_5 - b_4$ are in a G.P.

(D) $\frac{1}{b_3-b_2}$, $\frac{1}{b_4-b_3}$, $\frac{1}{b_5-b_4}$ are in an A.P. with common difference -2

Q114. JEE Main 2022 (Online) 25th June Morning Shift

The value of $\int_0^{\pi} \frac{e^{\cos x}\sin x}{(1+\cos^2 x)(e^{\cos x}+e^{-\cos x})}dx$ is equal to:

(A) $\frac{\pi^2}{4}$ (B) $\frac{\pi^2}{2}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Q115. JEE Main 2022 (Online) 24th June Evening Shift

The value of the integral

$\int_{-\pi/2}^{\pi/2} \frac{dx}{(1+e^x)(\sin^6 x+\cos^6 x)}$ is equal to

(A) 2π (B) 0 (C) π (D) $\frac{\pi}{2}$

Q116. JEE Main 2021 (Online) 1st September Evening Shift

Let f : R $\rightarrow$ R be a continuous function. Then $\lim\limits_{x\to\frac{\pi}{4}} \frac{\int_2^{\sec^2 x} f(x)\,dx}{x^2-\frac{\pi^2}{16}}$ is equal to :

(A) f (2) (B) 2f (2) (C) 2f $\left(\sqrt{2}\right)$ (D) 4f (2)

Q117. JEE Main 2021 (Online) 1st September Evening Shift

Let $J_{n,m} = \int_0^{\frac{1}{2}} \frac{x^n}{x^m-1}dx$, $\forall$ n > m and n, m $\in$ N. Consider a matrix $A = [a_{ij}]_{3\times 3}$ where $a_{ij} = \begin{cases} J_{6+i,3} - J_{i+3,3}, & i \le j \\ 0, & i > j \end{cases}$. Then $|adjA^{-1}|$ is :

(A) $(15)^2 \times 2^{42}$ (B) $(15)^2 \times 2^{34}$ (C) $(105)^2 \times 2^{38}$ (D) $(105)^2 \times 2^{36}$

Q118. JEE Main 2021 (Online) 1st September Evening Shift

The function f(x), that satisfies the condition $f(x) = x + \int_0^{\pi/2} \sin x.\cos y\, f(y)\,dy$, is :

(A) $x + \frac{2}{3}(\pi-2)\sin x$ (B) $x + (\pi+2)\sin x$ (C) $x + \frac{\pi}{2}\sin x$ (D) $x + (\pi-2)\sin x$

Q119. JEE Main 2021 (Online) 31st August Evening Shift

If [x] is the greatest integer $\le$ x, then

$\pi^2 \int_0^2 \left(\sin\frac{\pi x}{2}\right)(x-[x])^{[x]}dx$ is equal to :

(A) $2(\pi-1)$ (B) $4(\pi-1)$ (C) $4(\pi+1)$ (D) $2(\pi+1)$

Q120. JEE Main 2021 (Online) 31st August Morning Shift

Let f be a non-negative function in [0, 1] and twice differentiable in (0, 1). If $\int_0^x \sqrt{1-(f'(t))^2}dt = \int_0^x f(t)dt, 0 \le x \le 1$ and f(0) = 0, then $\lim\limits_{x\to 0} \frac{1}{x^2}\int_0^x f(t)dt$:

(A) equals 0 (B) equals 1 (C) does not exist (D) equals $\frac{1}{2}$

Q121. JEE Main 2021 (Online) 27th August Evening Shift

The value of the integral $\int\limits_0^1 \frac{\sqrt{x}dx}{(1+x)(1+3x)(3+x)}$ is :

(A) $\frac{\pi}{8}\left(1-\frac{\sqrt{3}}{2}\right)$ (B) $\frac{\pi}{4}\left(1-\frac{\sqrt{3}}{6}\right)$

(C) $\frac{\pi}{8}\left(1-\frac{\sqrt{3}}{6}\right)$ (D) $\frac{\pi}{4}\left(1-\frac{\sqrt{3}}{2}\right)$

Q122. JEE Main 2021 (Online) 27th August Morning Shift

$\int\limits_6^{16} \frac{\log_e x^2}{\log_e x^2+\log_e(x^2-44x+484)}dx$ is equal to :

(A) 6 (B) 8 (C) 5 (D) 10

Q123. JEE Main 2021 (Online) 26th August Evening Shift

If the value of the integral $\int\limits_0^5 \frac{x+[x]}{e^{x-[x]}}dx = \alpha e^{-1}+\beta$, where $\alpha, \beta \in$ R, $5\alpha + 6\beta = 0$, and [x] denotes the greatest integer less than or equal to x; then the value of $(\alpha+\beta)^2$ is equal to :

(A) 100 (B) 25 (C) 16 (D) 36

Q124. JEE Main 2021 (Online) 26th August Evening Shift

The value of $\int\limits_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1+\sin^2 x}{1+\pi^{\sin x}}\right)dx$ is

(A) $\frac{\pi}{2}$ (B) $\frac{5\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{2}$

Q125. JEE Main 2021 (Online) 26th August Morning Shift

The value of $\int\limits_{\frac{-1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left(\left(\frac{x+1}{x-1}\right)^2+\left(\frac{x-1}{x+1}\right)^2-2\right)^{\frac{1}{2}}dx$ is :

(A) $\log_e 4$ (B) $\log_e 16$

(C) $2\log_e 16$ (D) $4\log_e(3+2\sqrt{2})$

Q126. JEE Main 2021 (Online) 27th July Evening Shift

Let f : (a, b) $\rightarrow$ R be twice differentiable function such that $f(x) = \int_a^x g(t)dt$ for a differentiable function g(x). If f(x) = 0 has exactly five distinct roots in (a, b), then g(x)g'(x) = 0 has at least :

(A) twelve roots in (a, b)

(B) five roots in (a, b)

(C) seven roots in (a, b)

(D) three roots in (a, b)

Q127. JEE Main 2021 (Online) 27th July Morning Shift

The value of the definite integral

$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{dx}{(1+e^{x\cos x})(\sin^4 x+\cos^4 x)}$ is equal to :

(A) $-\frac{\pi}{2}$

(B) $\frac{\pi}{2\sqrt{2}}$

(C) $-\frac{\pi}{4}$

(D) $\frac{\pi}{\sqrt{2}}$

Q128. JEE Main 2021 (Online) 25th July Evening Shift

If $f(x) = \begin{cases} \int_0^x (5+|1-t|)dt, & x > 2 \\ 5x+1, & x \leq 2 \end{cases}$, then

(A) f(x) is not continuous at x = 2

(B) f(x) is everywhere differentiable

(C) f(x) is continuous but not differentiable at x = 2

(D) f(x) is not differentiable at x = 1

Q129. JEE Main 2021 (Online) 25th July Evening Shift

The value of the

integral $\int_{-1}^{1} \log\left(x+\sqrt{x^2+1}\right)dx$ is :

(A) 2

(B) 0

(C) -1

(D) 1

Q130. JEE Main 2021 (Online) 25th July Morning Shift

The value of the definite integral $\int_{\pi/24}^{5\pi/24} \frac{dx}{1+\sqrt[3]{\tan 2x}}$ is :

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{6}$

(C) $\frac{\pi}{12}$

(D) $\frac{\pi}{18}$

Q131. JEE Main 2021 (Online) 25th July Morning Shift

Let $f : [0, \infty) \to [0, \infty)$ be defined as $f(x) = \int_0^x [y]dy$

where [x] is the greatest integer less than or equal to x. Which of the following is true?

(A) f is continuous at every point in $[0, \infty)$ and differentiable except at the integer points.

(B) f is both continuous and differentiable except at the integer points in $[0, \infty)$.

(C) f is continuous everywhere except at the integer points in $[0, \infty)$.

(D) f is differentiable at every point in $[0, \infty)$.

Q132. JEE Main 2021 (Online) 22th July Evening Shift

If $\int_0^{100\pi} \frac{\sin^2 x}{e^{(\frac{x}{\pi} - [\frac{x}{\pi}])}} dx = \frac{\alpha\pi^3}{1+4\pi^2}, \alpha \in R$ where [x] is the greatest integer less than or equal to x, then the value of α is :

(A) 200 $(1 - e^{-1})$ (B) 100 $(1 - e)$ (C) 50 $(e - 1)$ (D) 150 $(e^{-1} - 1)$

Q133. JEE Main 2021 (Online) 20th July Evening Shift

If [x] denotes the greatest integer less than or equal to x, then the value of the integral $\int_{-\pi/2}^{\pi/2} [[x] - \sin x]dx$ is equal to :

(A) $-\pi$ (B) π (C) 0 (D) 1

Q134. JEE Main 2021 (Online) 20th July Evening Shift

If the real part of the complex number $(1 - \cos\theta + 2i\sin\theta)^{-1}$ is $\frac{1}{5}$ for $\theta \in (0, \pi)$, then the value of the integral $\int_0^\theta \sin x dx$ is equal to:

(A) 1 (B) 2 (C) -1 (D) 0

Q135. JEE Main 2021 (Online) 20th July Evening Shift

If $f : R \to R$ is given by $f(x) = x + 1$, then the value of $\lim_{n\to\infty} \frac{1}{n}\left[f(0) + f\left(\frac{5}{n}\right) + f\left(\frac{10}{n}\right) + \ldots\ldots + f\left(\frac{5(n-1)}{n}\right)\right]$ is :

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{1}{2}$ (D) $\frac{7}{2}$

Q136. JEE Main 2021 (Online) 20th July Evening Shift

Let $g(t) = \int_{-\pi/2}^{\pi/2} \cos\left(\frac{\pi}{4}t + f(x)\right)dx$, where $f(x) = \log_e\left(x + \sqrt{x^2 + 1}\right), x \in R$. Then which one of the following is correct?

(A) g(1) = g(0) (B) $\sqrt{2}g(1) = g(0)$ (C) $g(1) = \sqrt{2}g(0)$ (D) g(1) + g(0) = 0

Q137. JEE Main 2021 (Online) 20th July Morning Shift

Let a be a positive real number such that $\int_0^a e^{x-[x]}dx = 10e - 9$ where [x] is the greatest integer less than or equal to x. Then a is equal to:

(A) $10 - \log_e(1+e)$ (B) $10 + \log_e 2$ (C) $10 + \log_e 3$ (D) $10 + \log_e(1+e)$

Q138. JEE Main 2021 (Online) 20th July Morning Shift

The value of the integral $\int_{-1}^{1} \log_e(\sqrt{1-x} + \sqrt{1+x})dx$ is equal to:

(A) $\frac{1}{2}\log_e 2 + \frac{\pi}{4} - \frac{3}{2}$ (B) $2\log_e 2 + \frac{\pi}{4} - 1$

(C) $\log_e 2 + \frac{\pi}{2} - 1$ (D) $2\log_e 2 + \frac{\pi}{2} - \frac{1}{2}$

Q139. JEE Main 2021 (Online) 18th March Evening Shift

Let g(x) = $\int_0^x f(t)dt$, where f is continuous function in [0, 3] such that $\frac{1}{3} \leq$ f(t) ≤ 1 for all t$\in$ [0, 1] and 0 $\leq$ f(t) $\leq \frac{1}{2}$ for all t$\in$ (1, 3]. The largest possible interval in which g(3) lies is :

(A) $\left[-1, -\frac{1}{2}\right]$ (B) $\left[-\frac{3}{2}, -1\right]$

(C) [1, 3] (D) $\left[\frac{1}{3}, 2\right]$

Q140. JEE Main 2021 (Online) 17th March Evening Shift

Let f : R $\rightarrow$ R be defined as f(x) = e^{-x}sinx. If F : [0, 1] $\rightarrow$ R is a differentiable function with that F(x) = $\int_0^x f(t)dt$, then the value of $\int_0^1 (F'(x) + f(x))e^x dx$ lies in the interval

(A) $\left[\frac{331}{360}, \frac{334}{360}\right]$ (B) $\left[\frac{330}{360}, \frac{331}{360}\right]$

(C) $\left[\frac{335}{360}, \frac{336}{360}\right]$ (D) $\left[\frac{327}{360}, \frac{329}{360}\right]$

Q141. JEE Main 2021 (Online) 17th March Evening Shift

If the integral

$\int_0^{10} \frac{[\sin 2\pi x]}{e^{x-[x]}}dx = \alpha e^{-1} + \beta e^{-\frac{1}{2}} + \gamma$, where α, β, γ are integers and [x] denotes the greatest integer less than or equal to x, then the value of $\alpha + \beta + \gamma$ is equal to :

(A) 0 (B) 10 (C) 20 (D) 25

Q142. JEE Main 2021 (Online) 17th March Morning Shift

Which of the following statements is correct for the function g(α) for $\alpha \in$ R such that

$g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^{\alpha} x}{\cos^{\alpha} x + \sin^{\alpha} x} dx$

(A) $g(\alpha)$ is a strictly increasing function

(B) $g(\alpha)$ is an even function

(C) $g(\alpha)$ has an inflection point at $\alpha = -\frac{1}{2}$

(D) $g(\alpha)$ is a strictly decreasing function

Q143. JEE Main 2021 (Online) 16th March Evening Shift

Consider the integral
$I = \int_0^{10} \frac{[x]e^{[x]}}{e^{x-1}} dx$,
where [x] denotes the greatest integer less than or equal to x. Then the value of I is equal to :

(A) 45 (e $-$ 1) (B) 45 (e + 1) (C) 9 (e + 1) (D) 9 (e $-$ 1)

Q144. JEE Main 2021 (Online) 16th March Evening Shift

Let P(x) = x^2 + bx + c be a quadratic polynomial with real coefficients such that $\int_0^1 P(x)dx$ = 1 and P(x) leaves remainder 5 when it is divided by (x $-$ 2). Then the value of 9(b + c) is equal to :

(A) 9 (B) 11 (C) 7 (D) 15

Q145. JEE Main 2021 (Online) 26th February Evening Shift

Let $f(x) = \int_0^x e^t f(t)dt + e^x$ be a differentiable function for all x$\in$R. Then f(x) equals :

(A) $e^{(e^{x-1})}$ (B) $2e^{e^x} - 1$ (C) $2e^{e^x - 1} - 1$ (D) $e^{e^x} - 1$

Q146. JEE Main 2021 (Online) 26th February Evening Shift

For x > 0, if $f(x) = \int_1^x \frac{\log_e t}{(1+t)} dt$, then $f(e) + f\left(\frac{1}{e}\right)$ is equal to :

(A) $\frac{1}{2}$ (B) -1 (C) 0 (D) 1

Q147. JEE Main 2021 (Online) 26th February Morning Shift

The value of $\int_{-\pi/2}^{\pi/2} \frac{\cos^2 x}{1+3^x} dx$ is :

(A) 2π (B) $\frac{\pi}{2}$ (C) 4π (D) $\frac{\pi}{4}$

Q148. JEE Main 2021 (Online) 26th February Morning Shift

The value of $\sum_{n=1}^{100} \int_{n-1}^{n} e^{x-[x]}dx$, where [x] is the greatest integer $\leq$ x, is :

(A) 100e (B) 100(e $-$ 1) (C) 100(1 + e) (D) 100(1 $-$ e)

Q149. JEE Main 2021 (Online) 25th February Evening Shift

If $I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x \, dx$, then :

(A) $\frac{1}{I_2+I_4}, \frac{1}{I_3+I_5}, \frac{1}{I_4+I_6}$ are in A.P.

(B) $I_2 + I_4, I_3 + I_5, I_4 + I_6$ are in A.P.

(C) $\frac{1}{I_2+I_4}, \frac{1}{I_3+I_5}, \frac{1}{I_4+I_6}$ are in G.P.

(D) $I_2 + I_4, (I_3 + I_5)^2, I_4 + I_6$ are in G.P.

Q150. JEE Main 2021 (Online) 25th February Morning Shift

The value of $\int_{-1}^{1} x^2 e^{[x^3]}dx$, where [t] denotes the greatest integer $\leq$ t, is :

(A) $\frac{e+1}{3}$ (B) $\frac{e-1}{3e}$ (C) $\frac{1}{3e}$ (D) $\frac{e+1}{3e}$

Q151. JEE Main 2021 (Online) 24th February Evening Shift

The value of the integral, $\int_{1}^{3} [x^2 - 2x - 2]dx$, where [x] denotes the greatest integer less than or equal to x, is :

(A) -5 (B) $-\sqrt{2} - \sqrt{3} + 1$ (C) -4 (D) $-\sqrt{2} - \sqrt{3} - 1$

Q152. JEE Main 2021 (Online) 24th February Evening Shift

Let f(x) be a differentiable function defined on [0, 2] such that f'(x) = f'(2 $-$ x) for all x$\in$ (0, 2), f(0) = 1 and f(2) = e^2. Then the value of $\int_{0}^{2} f(x)dx$ is :

(A) 1 + e^2 (B) 2(1 + e^2) (C) 1 $-$ e^2 (D) 2(1 $-$ e^2)

Q153. JEE Main 2021 (Online) 24th February Evening Shift

Let f be a twice differentiable function defined on R such that f(0) = 1, f'(0) = 2 and f'(x) $\neq$ 0 for all x $\in$ R. If $\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix}$ = 0, for all x$\in$R, then the value of f(1) lies in the interval :

(A) (0, 3) (B) (9, 12) (C) (3, 6) (D) (6, 9)

Q154. JEE Main 2021 (Online) 24th February Morning Shift

$\lim\limits_{x\to 0}\dfrac{\int\limits_0^{x^2}\left(\sin\sqrt{t}\right)dt}{x^3}$ is equal to :

(A) $\frac{1}{15}$ (B) 0 (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

Q155. JEE Main 2020 (Online) 6th September Evening Slot

The integral $\int\limits_1^2 e^x.x^x\,(2+\log_e x)dx$ equals :

(A) e(4e + 1) (B) e(2e − 1) (C) e(4e − 1) (D) $4e^2 - 1$

Q156. JEE Main 2020 (Online) 6th September Morning Slot

$\lim\limits_{x\to 1}\left(\dfrac{\int\limits_0^{(x-1)^2} t\cos\left(t^2\right)dt}{(x-1)\sin(x-1)}\right)$

(A) is equal to 0 (B) is equal to $\frac{1}{2}$

(C) does not exist (D) is equal to $-\frac{1}{2}$

Q157. JEE Main 2020 (Online) 6th September Morning Slot

If $I_1 = \int\limits_0^1\left(1-x^{50}\right)^{100}dx$ and

$I_2 = \int\limits_0^1\left(1-x^{50}\right)^{101}dx$ such

that $I_2 = \alpha I_1$ then α equals to :

(A) $\frac{5051}{5050}$ (B) $\frac{5050}{5051}$ (C) $\frac{5050}{5049}$ (D) $\frac{5049}{5050}$

Q158. JEE Main 2020 (Online) 5th September Morning Slot

The value of $\int\limits_{\frac{-\pi}{2}}^{\frac{\pi}{2}}\frac{1}{1+e^{\sin x}}dx$ is:

(A) π (B) $\frac{3\pi}{2}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Q159. JEE Main 2020 (Online) 4th September Evening Slot

The integral

$\int\limits_{\frac{\pi}{6}}^{\frac{\pi}{3}}\tan^3 x.\sin^2 3x\left(2\sec^2 x.\sin^2 3x+3\tan x.\sin 6x\right)dx$

is equal to:

(A) $-\frac{1}{9}$ (B) $-\frac{1}{18}$ (C) $\frac{7}{18}$ (D) $\frac{9}{2}$

Q160. JEE Main 2020 (Online) 4th September Morning Slot

Let $f(x) = |x - 2|$ and g(x) = f(f(x)), $x \in [0, 4]$. Then $\int_0^3 (g(x) - f(x))dx$ is equal to:

(A) 1 (B) 0 (C) $\frac{1}{2}$ (D) $\frac{3}{2}$

Q161. JEE Main 2020 (Online) 3rd September Evening Slot

If the value of the integral

$\int_0^{\frac{1}{2}} \frac{x^2}{(1-x^2)^{\frac{3}{2}}}dx$

is $\frac{k}{6}$, then k is equal to :

(A) $2\sqrt{3} + \pi$ (B) $3\sqrt{2} - \pi$ (C) $3\sqrt{2} + \pi$ (D) $2\sqrt{3} - \pi$

Q162. JEE Main 2020 (Online) 3rd September Morning Slot

$\int_{-\pi}^{\pi} |\pi - |x||dx$ is equal to :

(A) π^2 (B) $2\pi^2$ (C) $\sqrt{2}\pi^2$ (D) $\frac{\pi^2}{2}$

Q163. JEE Main 2020 (Online) 9th January Evening Slot

Let a function f : [0, 5] $\rightarrow$ R be continuous, f(1) = 3 and F be defined as :

$F(x) = \int_1^x t^2 g(t)dt$, where $g(t) = \int_1^t f(u)du$

Then for the function F, the point x = 1 is :

(A) a point of inflection. (B) a point of local maxima.

(C) a point of local minima. (D) not a critical point.

Q164. JEE Main 2020 (Online) 9th January Morning Slot

The value of

$\int_0^{2\pi} \frac{x\sin^8 x}{\sin^8 x + \cos^8 x}dx$ is equal to :

(A) 4π (B) 2π (C) π^2 (D) $2\pi^2$

Q165. JEE Main 2020 (Online) 9th January Morning Slot

If for all real triplets (a, b, c), f(x) = a + bx + cx^2; then $\int_0^1 f(x)dx$ is equal to :

(A) $\frac{1}{6}\left\{f(0)+f(1)+4f\left(\frac{1}{2}\right)\right\}$

(B) $2\left\{3f(1)+2f\left(\frac{1}{2}\right)\right\}$

(C) $\frac{1}{3}\left\{f(0)+f\left(\frac{1}{2}\right)\right\}$

(D) $\frac{1}{2}\left\{f(1)+3f\left(\frac{1}{2}\right)\right\}$

Q166. JEE Main 2020 (Online) 8th January Evening Slot

If $I=\int_1^2 \frac{dx}{\sqrt{2x^3-9x^2+12x+4}}$, then :

(A) $\frac{1}{16}<I^2<\frac{1}{9}$

(B) $\frac{1}{8}<I^2<\frac{1}{4}$

(C) $\frac{1}{9}<I^2<\frac{1}{8}$

(D) $\frac{1}{6}<I^2<\frac{1}{2}$

Q167. JEE Main 2020 (Online) 8th January Evening Slot

$\lim_{x\to 0}\frac{\int_0^x t\sin(10t)dt}{x}$ is equal to

(A) $-\frac{1}{5}$

(B) $-\frac{1}{10}$

(C) 0

(D) $\frac{1}{10}$

Q168. JEE Main 2020 (Online) 7th January Evening Slot

The value of α for which

$4\alpha\int_{-1}^{2} e^{-\alpha|x|}dx=5$, is:

(A) $\log_e 2$

(B) $\log_e\sqrt{2}$

(C) $\log_e\left(\frac{4}{3}\right)$

(D) $\log_e\left(\frac{3}{2}\right)$

Q169. JEE Main 2020 (Online) 7th January Evening Slot

If θ_1 and θ_2 be respectively the smallest and the largest values of θ in (0, 2π) - {π} which satisfy the equation,

$2\cot^2\theta - \frac{5}{\sin\theta} + 4 = 0$, then

$\int_{\theta_1}^{\theta_2}\cos^2 3\theta d\theta$ is equal to :

(A) $\frac{\pi}{9}$

(B) $\frac{2\pi}{3}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{3}+\frac{1}{6}$

Q170. JEE Main 2020 (Online) 7th January Morning Slot

If f(a + b + 1 - x) = f(x), for all x, where a and b are fixed positive real numbers, then

$\frac{1}{a+b}\int_a^b x\left(f(x)+f(x+1)\right)dx$ is equal to:

(A) $\int_{a-1}^{b-1} f(x+1)dx$

(B) $\int_{a+1}^{b+1} f(x+1)dx$

(C) $\int_{a-1}^{b-1} f(x)dx$

(D) $\int_{a+1}^{b+1} f(x)dx$

Q171. JEE Main 2019 (Online) 12th April Evening Slot

A value of α such that $\int_{\alpha}^{\alpha+1} \frac{dx}{(x+\alpha)(x+\alpha+1)} = \log_e\left(\frac{9}{8}\right)$ is :

(A) 2 (B) - 2 (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

Q172. JEE Main 2019 (Online) 12th April Morning Slot

If $\int_0^{\frac{\pi}{2}} \frac{\cot x}{\cot x+\operatorname{cosec} x} dx$ = m(π + n), then m.n is equal to

(A) - 1 (B) 1 (C) $-\frac{1}{2}$ (D) $\frac{1}{2}$

Q173. JEE Main 2019 (Online) 12th April Morning Slot

Let f : R $\rightarrow$ R be a continuously differentiable function such that f(2) = 6 and f'(2) = $\frac{1}{48}$. If $\int_6^{f(x)} 4t^3 dt$ = (x - 2)g(x), then $\lim_{x\to 2} g(x)$ is equal to :

(A) 18 (B) 36 (C) 12 (D) 24

Q174. JEE Main 2019 (Online) 10th April Evening Slot

The integral $\int_{\pi/6}^{\pi/3} \sec^{2/3} x \operatorname{cosec}^{4/3} x dx$ is equal to :

(A) $3^{\frac{5}{3}} - 3^{\frac{1}{3}}$ (B) $3^{\frac{5}{6}} - 3^{\frac{2}{3}}$

(C) $3^{\frac{4}{3}} - 3^{\frac{1}{3}}$ (D) $3^{\frac{7}{6}} - 3^{\frac{5}{6}}$

Q175. JEE Main 2019 (Online) 10th April Morning Slot

The value of $\int_0^{2\pi} [\sin 2x\,(1+\cos 3x)] dx$,

where [t] denotes the greatest integer function is :

(A) 2π (B) π (C) -2π (D) $-\pi$

Q176. JEE Main 2019 (Online) 9th April Evening Slot

If f : R $\rightarrow$ R is a differentiable function and f(2) = 6,

then $\lim_{x\to 2} \frac{\int_6^{f(x)} 2t dt}{(x-2)}$ is :-

(A) 2f'(2) (B) 24f'(2) (C) 0 (D) 12f'(2)

Q177. JEE Main 2019 (Online) 9th April Evening Slot

The value of the integral $\int_0^1 x\cot^{-1}(1-x^2+x^4)dx$ is :-

(A) $\frac{\pi}{2}-\frac{1}{2}\log_e 2$ (B) $\frac{\pi}{4}-\log_e 2$

(C) $\frac{\pi}{4}-\frac{1}{2}\log_e 2$ (D) $\frac{\pi}{2}-\log_e 2$

Q178. JEE Main 2019 (Online) 9th April Morning Slot

The value of $\int_0^{\pi/2} \frac{\sin^3 x}{\sin x+\cos x}dx$ is

(A) $\frac{\pi-2}{8}$ (B) $\frac{\pi-2}{4}$ (C) $\frac{\pi-1}{2}$ (D) $\frac{\pi-1}{4}$

Q179. JEE Main 2019 (Online) 8th April Evening Slot

Let $f(x)=\int_0^x g(t)dt$ where g is a non-zero even function. If f(x + 5) = g(x), then $\int_0^x f(t)dt$ equals-

(A) $5\int_{x+5}^{5} g(t)dt$ (B) $\int_{x+5}^{5} g(t)dt$

(C) $\int_5^{x+5} g(t)dt$ (D) $2\int_5^{x+5} g(t)dt$

Q180. JEE Main 2019 (Online) 8th April Morning Slot

If $f(x)=\frac{2-x\cos x}{2+x\cos x}$ and g(x) = $\log_e$x, (x > 0) then the value of integral

$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} g\left(f\left(x\right)\right)dx$ is

(A) $\log_e 3$ (B) $\log_e 2$ (C) $\log_e 1$ (D) $\log_e e$

Q181. JEE Main 2019 (Online) 12th January Evening Slot

The integral $\int_1^e \left\{\left(\frac{x}{e}\right)^{2x}-\left(\frac{e}{x}\right)^{x}\right\}$ $\log_e$ x dx is equal to :

(A) $-\frac{1}{2}+\frac{1}{e}-\frac{1}{2e^2}$ (B) $\frac{3}{2}-e-\frac{1}{2e^2}$

(C) $\frac{1}{2}-e-\frac{1}{e^2}$ (D) $\frac{3}{2}-\frac{1}{e}-\frac{1}{2x^2}$

Q182. JEE Main 2019 (Online) 12th January Morning Slot

Let f and g be continuous functions on [0, a] such that f(x) = f(a − x) and g(x) + g(a − x) = 4, then $\int_0^a$ f(x) g(x) dx is equal to :

(A) $4\int_0^a$ f(x)dx (B) $-3\int_0^a$ f(x)dx

(C) $\int_0^a$ f(x)dx (D) $2\int_0^a$ f(x)dx

Q183. JEE Main 2019 (Online) 11th January Evening Slot

The integral $\int_{\pi/6}^{\pi/4} \frac{dx}{\sin 2x\left(\tan^5 x+\cot^5 x\right)}$ equals :

(A) $\frac{\pi}{40}$

(B) $\frac{1}{20}\tan^{-1}\left(\frac{1}{9\sqrt{3}}\right)$

(C) $\frac{1}{10}\left(\frac{\pi}{4}-\tan^{-1}\left(\frac{1}{9\sqrt{3}}\right)\right)$

(D) $\frac{1}{5}\left(\frac{\pi}{4}-\tan^{-1}\left(\frac{1}{3\sqrt{3}}\right)\right)$

Q184. JEE Main 2019 (Online) 11th January Morning Slot

The value of the integral $\int_{-2}^{2} \frac{\sin^2 x}{\left[\frac{x}{\pi}\right]+\frac{1}{2}}\,dx$ (where [x] denotes the greatest integer less than or equal to x) is

(A) 0 (B) 4 (C) 4– sin 4 (D) sin 4

Q185. JEE Main 2019 (Online) 10th January Evening Slot

The value of $\int_{-\pi/2}^{\pi/2} \frac{dx}{[x]+[\sin x]+4}$, where [t] denotes the greatest integer less than or equal to t, is

(A) $\frac{1}{12}(7\pi-5)$ (B) $\frac{1}{12}(7\pi+5)$

(C) $\frac{3}{10}(4\pi-3)$ (D) $\frac{3}{20}(4\pi-3)$

Q186. JEE Main 2019 (Online) 10th January Evening Slot

If $\int_0^x f(t)\,dt = x^2 + \int_x^1 t^2 f(t)\,dt$ then $f'\left(\frac{1}{2}\right)$ is -

(A) $\frac{18}{25}$ (B) $\frac{6}{25}$ (C) $\frac{24}{25}$ (D) $\frac{4}{5}$

Q187. JEE Main 2019 (Online) 10th January Morning Slot

Let $I=\int_a^b \left(x^4-2x^2\right)dx$. If I is minimum then the ordered pair (a, b) is -

(A) $\left(\sqrt{2},-\sqrt{2}\right)$ (B) $\left(0,\sqrt{2}\right)$

(C) $\left(-\sqrt{2},\sqrt{2}\right)$ (D) $\left(-\sqrt{2},0\right)$

Q188. JEE Main 2019 (Online) 9th January Evening Slot

Let f be a differentiable function from

R to **R** such that $|f(x) - f(y)| \leq 2|x-y|^{\frac{3}{2}}$,

for all $x, y \in$ **R**.

If $f(0) = 1$

then $\int_0^1 f^2(x)dx$ is equal to :

(A) 1 (B) 2 (C) $\frac{1}{2}$ (D) 0

Q189. JEE Main 2019 (Online) 9th January Evening Slot

If $\int_0^{\frac{\pi}{3}} \frac{\tan\theta}{\sqrt{2k\sec\theta}}d\theta = 1 - \frac{1}{\sqrt{2}}, (k > 0)$, then value of k is :

(A) 4 (B) $\frac{1}{2}$ (C) 1 (D) 2

Q190. JEE Main 2019 (Online) 9th January Morning Slot

The value of $\int_0^{\pi} |\cos x|^3 dx$ is :

(A) $\frac{4}{3}$ (B) $-\frac{4}{3}$ (C) 0 (D) $\frac{2}{3}$

Q191. JEE Main 2018 (Online) 16th April Morning Slot

If $f(x) = \int_0^x t(\sin x - \sin t)dt$ then :

(A) f'''(x) + f''(x) = sinx (B) f'''(x) + f''(x) $-$ f'(x) = cosx

(C) f'''(x) + f'(x) = cosx $-$ 2x sinx (D) f'''(x) $-$ f''(x) = cosx $-$ 2x sinx

Q192. JEE Main 2018 (Offline)

The value of $\int_{-\pi/2}^{\pi/2} \frac{\sin^2 x}{1+2^x}dx$ is

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{8}$ (C) $\frac{\pi}{2}$ (D) 4π

Q193. JEE Main 2018 (Online) 15th April Evening Slot

The value of integral $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{x}{1+\sin x}dx$ is :

(A) $\pi\sqrt{2}$ (B) $\pi\left(\sqrt{2}-1\right)$

(C) $\frac{\pi}{2}\left(\sqrt{2}+1\right)$ (D) $2\pi\left(\sqrt{2}-1\right)$

Q194. JEE Main 2018 (Online) 15th April Evening Slot

If $I_1 = \int_0^1 e^{-x}\cos^2 x\,dx$;

$I_2 = \int_0^1 e^{-x^2}\cos^2 x\,dx$ and

$I_3 = \int_0^1 e^{-x^3}dx$; then

(A) $I_2 > I_3 > I_1$ (B) $I_2 > I_1 > I_3$ (C) $I_3 > I_2 > I_1$ (D) $I_3 > I_1 > I_2$

Q195. JEE Main 2018 (Online) 15th April Morning Slot

The value of the integral

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^4 x\left(1+\log\left(\frac{2+\sin x}{2-\sin x}\right)\right)dx$ is :

(A) 0 (B) $\frac{3}{4}$ (C) $\frac{3}{8}\pi$ (D) $\frac{3}{16}\pi$

Q196. JEE Main 2017 (Online) 9th April Morning Slot

If $\int_1^2 \frac{dx}{(x^2-2x+4)^{\frac{3}{2}}} = \frac{k}{k+5}$, then k is equal to :

(A) 1 (B) 2 (C) 3 (D) 4

Q197. JEE Main 2017 (Online) 8th April Morning Slot

The integral $\int_{\frac{\pi}{12}}^{\frac{\pi}{4}} \frac{8\cos 2x}{(\tan x+\cot x)^3}\,dx$ equals :

(A) $\frac{15}{128}$ (B) $\frac{15}{64}$ (C) $\frac{13}{32}$ (D) $\frac{13}{256}$

Q198. JEE Main 2017 (Offline)

The integral $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{dx}{1+\cos x}$ is equal to

(A) 2 (B) 4 (C) -1 (D) -2

Q199. JEE Main 2016 (Online) 10th April Morning Slot

The value of the integral

$\int_4^{10} \frac{[x^2]dx}{[x^2-28x+196]+[x^2]}$,

where [x] denotes the greatest integer less than or equal to x, is :

(A) 6 (B) 3 (C) 7 (D) $\frac{1}{3}$

Q200. JEE Main 2016 (Online) 10th April Morning Slot

For x $\in$ **R**, x $\neq$ 0, if y(x) is a differentiable function such that

$x\int_1^x y(t)\,dt = (x+1)\int_1^x ty(t)\,dt$, then y (x) equals :

(where C is a constant.)

(A) $\frac{C}{x}e^{-\frac{1}{x}}$ (B) $\frac{C}{x^2}e^{-\frac{1}{x}}$

(C) $\frac{C}{x^3}e^{-\frac{1}{x}}$ (D) $Cx^3\,\frac{1}{e^x}$

Q201. JEE Main 2016 (Online) 9th April Morning Slot

If $2\int_0^1 \tan^{-1}x\,dx = \int_0^1 \cot^{-1}\left(1-x+x^2\right)dx,$

then $\int_0^1 \tan^{-1}\left(1-x+x^2\right)dx$ is equalto :

(A) log4 (B) $\frac{\pi}{2}$ + log2 (C) log2 (D) $\frac{\pi}{2}$ − log4

Q202. JEE Main 2015 (Offline)

The integral

$\int_2^4 \frac{\log x^2}{\log x^2+\log(36-12x+x^2)}dx$ is equal to :

(A) 1 (B) 6 (C) 2 (D) 4

Q203. JEE Main 2014 (Offline)

The integral $\int_0^{\pi}\sqrt{1+4\sin^2\frac{x}{2}-4\sin\frac{x}{2}}\,dx$ equals:

(A) $4\sqrt{3}-4$ (B) $4\sqrt{3}-4-\frac{\pi}{3}$

(C) $\pi-4$ (D) $\frac{2\pi}{3}-4-4\sqrt{3}$

Q204. JEE Main 2013 (Offline)

Statement-1 : The value of the integral $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$ is equal to $\pi/6$

Statement-2 : $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$.

(A) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.

(C) Statement- 1 is true; Statement-2 is False.

(D) Statement-1 is false; Statement-2 is true.

Q205. AIEEE 2011

The value of $\int_0^1 \frac{8\log(1+x)}{1+x^2}dx$ is

(A) $\frac{\pi}{8}\log 2$ (B) $\frac{\pi}{2}\log 2$ (C) $\log 2$ (D) $\pi \log 2$

Q206. AIEEE 2010

Let $p(x)$ be a function defined on R such that $p'(x) = p'(1-x)$, for all $x \in [0,1], p(0) = 1$ and $p(1) = 41$. Then $\int_0^1 p(x)dx$ equals :

(A) 21 (B) 41 (C) 42 (D) $\sqrt{41}$

Q207. AIEEE 2009

$\int_0^{\pi} [\cot x]dx$, where $[.]$ denotes the greatest integer function, is equal to:

(A) 1 (B) -1 (C) $-\frac{\pi}{2}$ (D) $\frac{\pi}{2}$

Q208. AIEEE 2007

Let $F(x) = f(x) + f\left(\frac{1}{x}\right)$, where $f(x) = \int_1^x \frac{\log t}{1+t}dt$, Then $F(e)$ equals

(A) 1 (B) 2 (C) $1/2$ (D) 0

Q209. AIEEE 2007

The solution for x of the equation $\int_{\sqrt{2}}^{x} \frac{dt}{t\sqrt{t^2-1}} = \frac{\pi}{2}$ is

(A) $\frac{\sqrt{3}}{2}$ (B) $2\sqrt{2}$ (C) 2 (D) None

Q210. AIEEE 2007

Let $I = \int_0^1 \frac{\sin x}{\sqrt{x}} dx$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}} dx$. Then which one of the following is true?

(A) $I > \frac{2}{3}$ and $J > 2$

(B) $I < \frac{2}{3}$ and $J < 2$

(C) $I < \frac{2}{3}$ and $J > 2$

(D) $I > \frac{2}{3}$ and $J < 2$

Q211. AIEEE 2006

$\int_0^{\pi} xf(\sin x)dx$ is equal to

(A) $\pi \int_0^{\pi} f(\cos x)dx$

(B) $\pi \int_0^{\pi} f(sinx)dx$

(C) $\frac{\pi}{2} \int_0^{\pi/2} f(sinx)dx$

(D) $\pi \int_0^{\pi/2} f(\cos x)dx$

Q212. AIEEE 2006

$\int_{-\frac{3\pi}{2}}^{-\frac{\pi}{2}} \left[(x+\pi)^3 + \cos^2(x+3\pi)\right] dx$ is equal to

(A) $\frac{\pi^4}{32}$

(B) $\frac{\pi^4}{32} + \frac{\pi}{2}$

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{4} - 1$

Q213. AIEEE 2006

The value of $\int_1^a [x]f'(x)dx, a > 1$ where $[x]$ denotes the greatest integer not exceeding x is

(A) $af(a) - \{f(1) + f(2) + \ldots\ldots\ldots\ldots f([a])\}$

(B) $[a]f(a) - \{f(1) + f(2) + \ldots\ldots\ldots\ldots f([a])\}$

(C) $[a]f([a]) - \{f(1) + f(2) + \ldots\ldots\ldots\ldots f(a)\}$

(D) $af([a]) - \{f(1) + f(2) + \ldots\ldots\ldots\ldots f(a)\}$

Q214. AIEEE 2005

Let $f : R \to R$ be a differentiable function having $f(2) = 6$, $f'(2) = \left(\frac{1}{48}\right)$. Then $\lim_{x \to 2} \int_6^{f(x)} \frac{4t^3}{x-2} dt$ equals :

(A) 24

(B) 36

(C) 12

(D) 18

Q215. AIEEE 2005

If $I_1 = \int_0^1 2^{x^2} dx, I_2 = \int_0^1 2^{x^3} dx$, $I_3 = \int_1^2 2^{x^2} dx$ and $I_4 = \int_1^2 2^{x^3} dx$ then

(A) $I_2 > I_1$

(B) $I_1 > I_2$

(C) $I_3 = I_4$

(D) $I_3 > I_4$

Q216. AIEEE 2005

The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x} dx,\ a > 0$, is

(A) $a\,\pi$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{a}$ (D) 2π

Q217. AIEEE 2005

The value of integral, $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x}+\sqrt{x}} dx$ is

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) 2 (D) 1

Q218. AIEEE 2004

The value of $\int_{-2}^{3} \left|1-x^2\right| dx$ is

(A) $\frac{1}{3}$ (B) $\frac{14}{3}$ (C) $\frac{7}{3}$ (D) $\frac{28}{3}$

Q219. AIEEE 2004

The value of $I = \int_0^{\pi/2} \frac{(\sin x+\cos x)^2}{\sqrt{1+\sin 2x}} dx$ is

(A) 3 (B) 1 (C) 2 (D) 0

Q220. AIEEE 2004

If $\int_0^{\pi} x f(\sin x) dx = A \int_0^{\pi/2} f(\sin x) dx$, then A is

(A) 2π (B) π (C) $\frac{\pi}{4}$ (D) 0

Q221. AIEEE 2004

If $f(x) = \frac{e^x}{1+e^x}, I_1 = \int_{f(-a)}^{f(a)} x g\{x(1-x)\} dx$

and $I_2 = \int_{f(-a)}^{f(a)} g\{x(1-x)\} dx$, then the value of $\frac{I_2}{I_1}$ is

(A) 1 (B) -3 (C) -1 (D) 2

Q222. AIEEE 2003

If $f(y) = e^y, g(y) = y; y > 0$ and

$F(t) = \int_0^t f(t-y) g(y) dy$, then :

(A) $F(t) = te^{-t}$ (B) $F(t) = 1t - te^{-1}(1+t)$

(C) $F(t) = e^t - (1+t)$ (D) $F(t) = te^t$.

Q223. AIEEE 2003

If $f(a+b-x)=f(x)$ then $\int_a^b xf(x)dx$ is equal to

(A) $\frac{a+b}{2}\int_a^b f(a+b+x)dx$

(B) $\frac{a+b}{2}\int_a^b f(b-x)dx$

(C) $\frac{a+b}{2}\int_a^b f(x)dx$

(D) $\frac{b-a}{2}\int_a^b f(x)dx$

Q224. AIEEE 2003

Let $f(x)$ be a function satisfying $f'(x)=f(x)$ with $f(0)=1$ and $g(x)$ be a function that satisfies $f(x)+g(x)=x^2$. Then the value of the integral $\int_0^1 f(x)g(x)dx$, is

(A) $e+\frac{e^2}{2}+\frac{5}{2}$

(B) $e-\frac{e^2}{2}-\frac{5}{2}$

(C) $e+\frac{e^2}{2}-\frac{3}{2}$

(D) $e-\frac{e^2}{2}-\frac{3}{2}$

Q225. AIEEE 2003

The value of the integral $I=\int_0^1 x(1-x)^n dx$ is

(A) $\frac{1}{n+1}+\frac{1}{n+2}$

(B) $\frac{1}{n+1}$

(C) $\frac{1}{n+2}$

(D) $\frac{1}{n+1}-\frac{1}{n+2}$

Q226. AIEEE 2003

The value of $\lim_{x\to 0}\frac{\int_0^{x^2}\sec^2 t dt}{x\sin x}$ is

(A) 0 (B) 3 (C) 2 (D) 1

Q227. AIEEE 2002

$\int_0^{10\pi}|\sin x|dx$ is

(A) 20 (B) 8 (C) 10 (D) 18

Q228. AIEEE 2002

$I_n=\int_0^{\pi/4}\tan^n x\,dx$ then $\lim_{n\to\infty} n\left[I_n+I_{n+2}\right]$ equals

(A) $\frac{1}{2}$ (B) 1 (C) ∞ (D) zero

Q229. AIEEE 2002

$\int_0^2 [x^2]dx$ is

(A) $2-\sqrt{2}$ (B) $2+\sqrt{2}$ (C) $\sqrt{2}-1$ (D) $-\sqrt{2}-\sqrt{3}+5$

Q230. AIEEE 2002

$\int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x}dx$ is

(A) $\frac{\pi^2}{4}$ (B) π^2 (C) zero (D) $\frac{\pi}{2}$

Q231. AIEEE 2002

If $y=f(x)$ makes $+ve$ intercept of 2 and 0 unit on x and y axes and encloses an area of $3/4$ square unit with the axes then $\int_0^2 xf'(x)dx$ is

(A) $3/2$ (B) 1 (C) $5/4$ (D) $-3/4$

Numerical 77 questions

Q232. JEE Main 2026 (Online) 8th April Evening Shift

If $\int_{\pi/6}^{\pi/4} \left(\cot\left(x-\frac{\pi}{3}\right)\cot\left(x+\frac{\pi}{3}\right)+1\right)dx = \alpha\log_e(\sqrt{3}-1)$, then $9\alpha^2$ is equal to ____.

Q233. JEE Main 2026 (Online) 2nd April Morning Shift

If $\alpha = \int_0^{2\sqrt{3}} \log_2(x^2+4)\,dx + \int_2^4 \sqrt{2^x-4}\,dx$, then α^2 is equal to _____.

Q234. JEE Main 2026 (Online) 28th January Evening Shift

Let f be a differentiable function satisfying $f(x)=1-2x+\int_0^x e^{(x-t)}f(t)\,dt, x\in\mathbb{R}$ and let

$g(x)=\int_0^x (f(t)+2)^{15}(t-4)^6(t+12)^{17}\,dt, x\in\mathbb{R}$.

If p and q are respectively the points of local minima and local maxima of g, then the value of $|p+q|$ is equal to _____.

Q235. JEE Main 2026 (Online) 28th January Morning Shift

The value of $\sum_{r=1}^{20}\left(\left|\sqrt{\pi\left(\int_0^r x|\sin\pi x|dx\right)}\right|\right)$ is ____

Q236. JEE Main 2026 (Online) 24th January Evening Shift

If $f(x)$ satisfies the relation $f(x) = e^x + \int_0^1 (y + xe^x) f(y) dy$, then $e + f(0)$ is equal to ____ .

Q237. JEE Main 2026 (Online) 24th January Morning Shift

Let a differentiable function f satisfy the equation $\int_0^{36} f\left(\frac{tx}{36}\right) dt = 4\alpha f(x)$. If $y = f(x)$ is a standard parabola passing through the points $(2, 1)$ and $(-4, \beta)$, then β^α is equal to ____ .

Q238. JEE Main 2026 (Online) 23rd January Evening Shift

The number of elements in the set $S = \{x : x \in [0, 100] \text{ and } \int_0^x t^2 \sin(x - t) dt = x^2\}$ is ____

Q239. JEE Main 2026 (Online) 22nd January Evening Shift

Let [.] be the greatest integer function. If $\alpha = \int_0^{64} \left(x^{1/3} - \left[x^{1/3}\right]\right) dx$, then $\frac{1}{\pi} \int_0^{\alpha\pi} \left(\frac{\sin^2\theta}{\sin^6\theta + \cos^6\theta}\right) d\theta$ is equal to ____ .

Q240. JEE Main 2026 (Online) 21st January Evening Shift

Let $[\cdot]$ denote the greatest integer function and $f(x) = \lim_{n\to\infty} \frac{1}{n^3} \sum_{k=1}^{n} \left[\frac{k^2}{3^x}\right]$. Then $12 \sum_{j=1}^{\infty} f(j)$ is equal to _____.

Q241. JEE Main 2026 (Online) 21st January Evening Shift

If $\int_0^1 4 \cot^{-1}\left(1 - 2x + 4x^2\right) dx = a \tan^{-1}(2) - b \log_e(5)$, where $a, b \in N$, then $(2a + b)$ is equal to ____ .

Q242. JEE Main 2026 (Online) 21st January Morning Shift

$6 \int_0^{\pi} |(\sin 3x + \sin 2x + \sin x)| dx$ is equal to ____ .

Q243. JEE Main 2025 (Online) 2nd April Morning Shift

Let [.] denote the greatest integer function. If $\int_0^{e^3} \left[\frac{1}{e^{x-1}}\right] dx = \alpha - \log_e 2$, then α^3 is equal to _____.

Q244. JEE Main 2025 (Online) 29th January Evening Shift

If $24\int_0^{\frac{\pi}{4}}\left(\sin\left|4x-\frac{\pi}{12}\right|+[2\sin x]\right)dx=2\pi+\alpha$, where $[\cdot]$ denotes the greatest integer function, then α is equal to _____.

Q245. JEE Main 2025 (Online) 29th January Evening Shift

If $\lim\limits_{t\to 0}\left(\int_0^1(3x+5)^t dx\right)^{\frac{1}{t}}=\frac{\alpha}{5e}\left(\frac{8}{5}\right)^{\frac{2}{5}}$, then α is equal to _____.

Q246. JEE Main 2025 (Online) 29th January Morning Shift

Let $f:(0,\infty)\to\mathbf{R}$ be a twice differentiable function. If for some $a\neq 0, \int_0^1 f(\lambda x)\mathrm{d}\lambda=af(x), f(1)=1$ and $f(16)=\frac{1}{8}$, then $16-f'\left(\frac{1}{16}\right)$ is equal to _____.

Q247. JEE Main 2024 (Online) 9th April Morning Shift

Let $\lim\limits_{n\to\infty}\left(\frac{n}{\sqrt{n^4+1}}-\frac{2n}{(n^2+1)\sqrt{n^4+1}}+\frac{n}{\sqrt{n^4+16}}-\frac{8n}{(n^2+4)\sqrt{n^4+16}}+\ldots+\frac{n}{\sqrt{n^4+n^4}}-\frac{2n\cdot n^2}{(n^2+n^2)\sqrt{n^4+n^4}}\right)$ be $\frac{\pi}{k}$, using only the principal values of the inverse trigonometric functions. Then k^2 is equal to _____.

Q248. JEE Main 2024 (Online) 6th April Evening Shift

Let $[t]$ denote the largest integer less than or equal to t. If $\int_0^3\left([x^2]+\left[\frac{x^2}{2}\right]\right)\mathrm{d}x=\mathrm{a}+\mathrm{b}\sqrt{2}-\sqrt{3}-\sqrt{5}+\mathrm{c}\sqrt{6}-\sqrt{7}$, where $\mathrm{a},\mathrm{b},\mathrm{c}\in\mathbf{Z}$, then $\mathrm{a}+\mathrm{b}+\mathrm{c}$ is equal to _____.

Q249. JEE Main 2024 (Online) 6th April Morning Shift

Let $r_k=\frac{\int_0^1(1-x^7)^k dx}{\int_0^1(1-x^7)^{k+1}dx}, k\in\mathbb{N}$. Then the value of $\sum\limits_{k=1}^{10}\frac{1}{7(r_k-1)}$ is equal to _____.

Q250. JEE Main 2024 (Online) 5th April Evening Shift

If $f(t)=\int_0^{\pi}\frac{2x\,\mathrm{d}x}{1-\cos^2\mathrm{t}\sin^2 x}, 0<\mathrm{t}<\pi$, then the value of $\int_0^{\frac{\pi}{2}}\frac{\pi^2\mathrm{dt}}{f(\mathrm{t})}$ equals _____.

Q251. JEE Main 2024 (Online) 4th April Morning Shift

If the shortest distance between the lines $\frac{x+2}{2}=\frac{y+3}{3}=\frac{z-5}{4}$ and $\frac{x-3}{1}=\frac{y-2}{-3}=\frac{z+4}{2}$ is $\frac{38}{3\sqrt{5}}\mathrm{k}$, and $\int_0^{\mathrm{k}}[x^2]\mathrm{d}x=\alpha-\sqrt{\alpha}$, where $[x]$ denotes the greatest integer function, then $6\alpha^3$ is equal to _____.

Q252. JEE Main 2024 (Online) 4th April Morning Shift

If $\int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1+\sin x \cos x}\, dx = \frac{1}{a}\log_e\left(\frac{a}{3}\right) + \frac{\pi}{b\sqrt{3}}$, where $a, b \in N$, then $a + b$ is equal to _____.

Q253. JEE Main 2024 (Online) 1st February Evening Shift

Let $f : (0, \infty) \to \mathbf{R}$ and $F(x) = \int_0^x t f(t) dt$. If $F\left(x^2\right) = x^4 + x^5$, then $\sum_{r=1}^{12} f\left(r^2\right)$ is equal to _____.

Q254. JEE Main 2024 (Online) 1st February Morning Shift

If $\int_{-\pi/2}^{\pi/2} \frac{8\sqrt{2}\cos x\, dx}{(1+e^{\sin x})(1+\sin^4 x)} = \alpha\pi + \beta\log_e(3+2\sqrt{2})$, where α, β are integers, then $\alpha^2 + \beta^2$ equals :

Q255. JEE Main 2024 (Online) 31st January Evening Shift

$\left|\frac{120}{\pi^3}\int_0^{\pi} \frac{x^2 \sin x \cos x}{\sin^4 x + \cos^4 x} dx\right|$ is equal to _____.

Q256. JEE Main 2024 (Online) 31st January Morning Shift

If the integral $525\int_0^{\frac{\pi}{2}} \sin 2x \cos^{\frac{11}{2}} x\left(1 + \mathrm{Cos}^{\frac{5}{2}} x\right)^{\frac{1}{2}} dx$ is equal to $(n\sqrt{2} - 64)$, then n is equal to _____.

Q257. JEE Main 2024 (Online) 31st January Morning Shift

Let $S = (-1, \infty)$ and $f : S \to \mathbb{R}$ be defined as

$f(x) = \int_{-1}^{x} \left(e^t - 1\right)^{11}(2t-1)^5(t-2)^7(t-3)^{12}(2t-10)^{61} dt,$

Let $p =$ Sum of squares of the values of x, where $f(x)$ attains local maxima on S, and $q =$ Sum of the values of x, where $f(x)$ attains local minima on S. Then, the value of $p^2 + 2q$ is _____.

Q258. JEE Main 2024 (Online) 31st January Morning Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \frac{4^x}{4^x+2}$ and

$M = \int_{f(a)}^{f(1-a)} x \sin^4(x(1-x)) dx, N = \int_{f(a)}^{f(1-a)} \sin^4(x(1-x)) dx; a \neq \frac{1}{2}$. If $\alpha M = \beta N, \alpha, \beta \in \mathbb{N}$, then the least value of $\alpha^2 + \beta^2$ is equal to _____.

Q259. JEE Main 2024 (Online) 30th January Morning Shift

The value of $9\int_0^9\left[\sqrt{\frac{10x}{x+1}}\right]\mathrm{d}x$, where $[t]$ denotes the greatest integer less than or equal to t, is

Q260. JEE Main 2024 (Online) 29th January Evening Shift

Let the slope of the line $45x+5y+3=0$ be $27r_1+\frac{9r_2}{2}$ for some $r_1, r_2 \in \mathbb{R}$. Then $\lim\limits_{x\to 3}\left(\int_3^x \frac{8t^2}{\frac{3r_2x}{2}-r_2x^2-r_1x^3-3x}dt\right)$ is equal to _____.

Q261. JEE Main 2024 (Online) 29th January Evening Shift

If $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\sqrt{1-\sin 2x}dx=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}$, where α, β and γ are rational numbers, then $3\alpha+4\beta-\gamma$ is equal to _____.

Q262. JEE Main 2024 (Online) 27th January Evening Shift

Let $f(x)=\int_0^x g(t)\log_e\left(\frac{1-t}{1+t}\right)\mathrm{dt}$, where g is a continuous odd function. If $\int_{-\pi/2}^{\pi/2}\left(f(x)+\frac{x^2\cos x}{1+e^x}\right)\mathrm{d}x=\left(\frac{\pi}{\alpha}\right)^2-\alpha$, then α is equal to _____.

Q263. JEE Main 2023 (Online) 13th April Evening Shift

Let $f_n=\int_0^{\frac{\pi}{2}}\left(\sum_{k=1}^n \sin^{k-1}x\right)\left(\sum_{k=1}^n (2k-1)\sin^{k-1}x\right)\cos x dx, n\in\mathbb{N}$. Then $f_{21}-f_{20}$ is equal to _____

Q264. JEE Main 2023 (Online) 13th April Morning Shift

Let for $x\in\mathbb{R}, S_0(x)=x, S_k(x)=C_kx+k\int_0^x S_{k-1}(t)dt$, where

$C_0=1, C_k=1-\int_0^1 S_{k-1}(x)dx, k=1,2,3,\ldots$ Then $S_2(3)+6C_3$ is equal to _____.

Q265. JEE Main 2023 (Online) 12th April Morning Shift

If $\int_{-0.15}^{0.15}\left|100x^2-1\right|dx=\frac{k}{3000}$, then k is equal to _____.

Q266. JEE Main 2023 (Online) 11th April Morning Shift

For $m, n>0$, let $\alpha(m,n)=\int_0^2 t^m(1+3t)^n dt$. If $11\alpha(10,6)+18\alpha(11,5)=p(14)^6$, then p is equal to _____.

Q267. JEE Main 2023 (Online) 8th April Evening Shift

Let $[t]$ denote the greatest integer function. If $\int_0^{2.4} [x^2]dx = \alpha + \beta\sqrt{2} + \gamma\sqrt{3} + \delta\sqrt{5}$, then $\alpha + \beta + \gamma + \delta$ is equal to _____.

Q268. JEE Main 2023 (Online) 8th April Morning Shift

Let $[t]$ denote the greatest integer $\leq t$. Then $\frac{2}{\pi}\int_{\pi/6}^{5\pi/6} (8[\operatorname{cosec} x] - 5[\cot x])dx$ is equal to _____.

Q269. JEE Main 2023 (Online) 1st February Evening Shift

If $\int_0^{\pi} \frac{5^{\cos x}(1+\cos x \cos 3x+\cos^2 x+\cos^3 x \cos 3x)dx}{1+5^{\cos x}} = \frac{k\pi}{16}$, then k is equal to _____.

Q270. JEE Main 2023 (Online) 1st February Morning Shift

If $\int_0^1 \left(x^{21} + x^{14} + x^7\right)\left(2x^{14} + 3x^7 + 6\right)^{1/7} dx = \frac{1}{l}(11)^{m/n}$ where $l, m, n \in \mathbb{N}$, m and n are coprime then $l + m + n$ is equal to _____.

Q271. JEE Main 2023 (Online) 1st February Morning Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $f'(x) + f(x) = \int_0^2 f(t)dt$. If $f(0) = e^{-2}$, then $2f(0) - f(2)$ is equal to _____.

Q272. JEE Main 2023 (Online) 30th January Morning Shift

$\lim_{x\to 0} \frac{48}{x^4}\int_0^x \frac{t^3}{t^6+1}\,dt$ is equal to _____.

Q273. JEE Main 2023 (Online) 25th January Evening Shift

If $\int_{\frac{1}{3}}^{3} |\log_e x|dx = \frac{m}{n}\log_e\left(\frac{n^2}{e}\right)$, where m and n are coprime natural numbers, then $m^2 + n^2 - 5$ is equal to _____.

Q274. JEE Main 2023 (Online) 24th January Evening Shift

Let f be a differentiable function defined on $\left[0, \frac{\pi}{2}\right]$ such that $f(x) > 0$ and $f(x) + \int_0^x f(t)\sqrt{1 - (\log_e f(t))^2}dt = e, \forall x \in \left[0, \frac{\pi}{2}\right]$. Then $\left(6\log_e f\left(\frac{\pi}{6}\right)\right)^2$ is equal to _____.

Q275. JEE Main 2023 (Online) 24th January Morning Shift

The value of $12\int_0^3 |x^2-3x+2|dx$ is _____

Q276. JEE Main 2023 (Online) 24th January Morning Shift

The value of $\frac{8}{\pi}\int_0^{\frac{\pi}{2}} \frac{(\cos x)^{2023}}{(\sin x)^{2023}+(\cos x)^{2023}}dx$ is _____

Q277. JEE Main 2022 (Online) 28th July Evening Shift

The value of the integral $\int_0^{\frac{\pi}{2}} 60\frac{\sin(6x)}{\sin x}dx$ is equal to _____.

Q278. JEE Main 2022 (Online) 28th July Morning Shift

If $\int_0^{\sqrt{3}} \frac{15x^3}{\sqrt{1+x^2+\sqrt{(1+x^2)^3}}}\,\mathrm{d}x = \alpha\sqrt{2}+\beta\sqrt{3}$, where α, β are integers, then $\alpha+\beta$ is equal to _____.

Q279. JEE Main 2022 (Online) 27th July Evening Shift

Let $f(x) = \min\{[x-1],[x-2],\ldots,[x-10]\}$ where [t] denotes the greatest integer $\leq$ t. Then $\int_0^{10} f(x)\mathrm{d}x + \int_0^{10}(f(x))^2\,\mathrm{d}x + \int_0^{10}|f(x)|\mathrm{d}x$ is equal to _____.

Q280. JEE Main 2022 (Online) 27th July Evening Shift

Let f be a differentiable function satisfying $f(x) = \frac{2}{\sqrt{3}}\int_0^{\sqrt{3}} f\left(\frac{\lambda^2 x}{3}\right)\mathrm{d}\lambda, x>0$ and $f(1)=\sqrt{3}$. If $y=f(x)$ passes through the point $(\alpha, 6)$, then α is equal to _____.

Q281. JEE Main 2022 (Online) 26th July Morning Shift

If $\mathrm{n}(2\mathrm{n}+1)\int_0^1 (1-x^{\mathrm{n}})^{2\mathrm{n}}\mathrm{d}x = 1177\int_0^1 (1-x^{\mathrm{n}})^{2\mathrm{n}+1}\,\mathrm{d}x$, then $\mathrm{n}\in\mathbf{N}$ is equal to _____.

Q282. JEE Main 2022 (Online) 25th July Evening Shift

Let f be a twice differentiable function on $\mathbb{R}$. If $f'(0)=4$ and $f(x)+\int_0^x (x-t)f'(t)dt = \left(e^{2x}+e^{-2x}\right)\cos 2x + \frac{2}{a}x$, then $(2a+1)^5 a^2$ is equal to _____.

Q283. JEE Main 2022 (Online) 25th July Evening Shift

Let $a_n = \int_{-1}^{n} \left(1 + \frac{x}{2} + \frac{x^2}{3} + \ldots\ldots + \frac{x^{n-1}}{n}\right) dx$ for every n ∈ N. Then the sum of all the elements of the set {n ∈ N : a_n ∈ (2, 30)} is _____.

Q284. JEE Main 2022 (Online) 30th June Morning Shift

Let $f(t) = \int_{0}^{t} e^{x^3} \left(\frac{x^8}{(x^6+2x^3+2)^2}\right) dx$. If $f(1) + f'(1) = \alpha e - \frac{1}{6}$, then the value of 150α is equal to _____.

Q285. JEE Main 2022 (Online) 26th June Evening Shift

The integral $\frac{24}{\pi} \int_0^{\sqrt{2}} \frac{(2-x^2)dx}{(2+x^2)\sqrt{4+x^4}}$ is equal to _____.

Q286. JEE Main 2022 (Online) 26th June Morning Shift

Let f(x) = max {|x + 1|, |x + 2|, ..., |x + 5|}. Then $\int_{-6}^{0} f(x)dx$ is equal to _____.

Q287. JEE Main 2022 (Online) 26th June Morning Shift

The value of the integral

$\frac{48}{\pi^4} \int_0^{\pi} \left(\frac{3\pi x^2}{2} - x^3\right) \frac{\sin x}{1+\cos^2 x} dx$ is equal to _____.

Q288. JEE Main 2022 (Online) 25th June Evening Shift

The value of b > 3 for which $12 \int_3^b \frac{1}{(x^2-1)(x^2-4)} dx = \log_e\left(\frac{49}{40}\right)$, is equal to _____.

Q289. JEE Main 2022 (Online) 24th June Morning Shift

Let $f(\theta) = \sin\theta + \int_{-\pi/2}^{\pi/2} (\sin\theta + t\cos\theta) f(t)dt$. Then the value of $\left|\int_0^{\pi/2} f(\theta)d\theta\right|$ is _____.

Q290. JEE Main 2022 (Online) 24th June Morning Shift

Let $\underset{0 \le x \le 2}{Max} \left\{\frac{9-x^2}{5-x}\right\} = \alpha$ and $\underset{0 \le x \le 2}{Min} \left\{\frac{9-x^2}{5-x}\right\} = \beta$.

If $\int_{\beta-\frac{8}{3}}^{2\alpha-1} Max\left\{\frac{9-x^2}{5-x}, x\right\} dx = \alpha_1 + \alpha_2 \log_e\left(\frac{8}{15}\right)$ then $\alpha_1 + \alpha_2$ is equal to _____.

Q291. JEE Main 2021 (Online) 31st August Morning Shift

Let [t] denote the greatest integer $\leq$ t. Then the value of

8. $\int_{-\frac{1}{2}}^{1}([2x]+|x|)dx$ is _____.

Q292. JEE Main 2021 (Online) 31st August Morning Shift

If $x\phi(x)=\int_{5}^{x}(3t^2-2\phi'(t))dt$, x > -2, and $\phi(0)$ = 4, then $\phi(2)$ is _____.

Q293. JEE Main 2021 (Online) 27th July Evening Shift

If $\int_0^{\pi}(\sin^3 x)e^{-\sin^2 x}dx=\alpha-\frac{\beta}{e}\int_0^1\sqrt{t}e^t dt$, then $\alpha+\beta$ is equal to _____.

Q294. JEE Main 2021 (Online) 27th July Morning Shift

Let the domain of the function

$f(x)=\log_4\left(\log_5\left(\log_3(18x-x^2-77)\right)\right)$ be (a, b). Then the value of the integral $\int_a^b\frac{\sin^3 x}{\sin^3 x+\sin^3(a+b-x)}dx$ is equal to _____.

Q295. JEE Main 2021 (Online) 27th July Morning Shift

Let $F:[3,5]\to R$ be a twice differentiable function on (3, 5) such that

$F(x)=e^{-x}\int_3^x(3t^2+2t+4F'(t))dt$. If $F'(4)=\frac{\alpha e^{\beta}-224}{(e^{\beta}-4)^2}$, then $\alpha+\beta$ is equal to _____.

Q296. JEE Main 2021 (Online) 18th March Evening Shift

Let P(x) be a real polynomial of degree 3 which vanishes at x = -3. Let P(x) have local minima at x = 1, local maxima at x = -1 and $\int_{-1}^{1}P(x)dx$ = 18, then the sum of all the coefficients of the polynomial P(x) is equal to _____.

Q297. JEE Main 2021 (Online) 18th March Morning Shift

Let f(x) and g(x) be two functions satisfying $f(x^2)+g(4-x)=4x^3$ and $g(4-x)+g(x)=0$, then the value of $\int_{-4}^{4}f(x^2)dx$ is

Q298. JEE Main 2021 (Online) 17th March Evening Shift

Let $I_n = \int_1^e x^{19}(\log|x|)^n dx$, where n∈N. If (20)$I_{10}$ = αI_9 + βI_8, for natural numbers α and β, then $\alpha - \beta$ equals to _____.

Q299. JEE Main 2021 (Online) 17th March Morning Shift

If [.] represents the greatest integer function, then the value of

$\left|\int_0^{\sqrt{\frac{\pi}{2}}} \left[[x^2] - \cos x\right] dx\right|$ is _____.

Q300. JEE Main 2021 (Online) 16th March Morning Shift

Let f : R $\rightarrow$ R be a continuous function such that f(x) + f(x + 1) = 2, for all x∈R.

If $I_1 = \int_0^8 f(x)dx$ and $I_2 = \int_{-1}^3 f(x)dx$, then the value of I_1 + 2I_2 is equal to _____.

Q301. JEE Main 2021 (Online) 16th March Morning Shift

If the normal to the curve y(x) = $\int_0^x (2t^2 - 15t + 10)dt$ at a point (a, b) is parallel to the line x + 3y = -5, a > 1, then the value of | a + 6b | is equal to _____.

Q302. JEE Main 2021 (Online) 26th February Evening Shift

If $I_{m,n} = \int_0^1 x^{m-1}(1-x)^{n-1}dx$, for m, $n \geq 1$, and

$\int_0^1 \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}dx = \alpha I_{m,n} \alpha \in R$, then α equals _____.

Q303. JEE Main 2021 (Online) 26th February Morning Shift

The value of the integral $\int_0^{\pi} |\sin 2x|dx$ is _____.

Q304. JEE Main 2021 (Online) 25th February Evening Shift

The value of $\int_{-2}^2 |3x^2 - 3x - 6|dx$ is _____.

Q305. JEE Main 2021 (Online) 24th February Morning Shift

If $\int_{-a}^{a}(|x|+|x-2|)dx = 22$, (a > 2) and [x] denotes the greatest integer $\leq$ x, then $\int_{a}^{-a}(x+[x])dx$ is equal to _____.

Q306. JEE Main 2020 (Online) 4th September Evening Slot

Let {x} and [x] denote the fractional part of x and the greatest integer $\leq$ x respectively of a real number x. If $\int_0^n \{x\}dx$, $\int_0^n [x]dx$ and $10(n^2 - n)$, $(n \in N, n > 1)$ are three consecutive terms of a G.P., then n is equal to_____.

Q307. JEE Main 2020 (Online) 2nd September Evening Slot

Let [t] denote the greatest integer less than or equal to t.

Then the value of $\int_1^2 |2x - [3x]|dx$ is _____.

Q308. JEE Main 2020 (Online) 2nd September Morning Slot

The integral $\int_0^2 ||x-1|-x|dx$

is equal to_____.

MCQ (More than One Correct Answer) 1 questions

Q309. AIEEE 2012

If $g(x) = \int_0^x \cos 4t\, dt$, then $g(x+\pi)$ equals

(A) $\frac{g(x)}{g(\pi)}$

(B) $g(x) + g(\pi)$

(C) $g(x) - g(\pi)$

(D) $g(x).g(\pi)$

Out of syllabus 22 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q310. JEE Main 2023 (Online) 13th April Morning Shift · MCQ (Single Correct Answer)

Among

(S1): $\lim\limits_{n\to\infty}\frac{1}{n^2}(2+4+6+\ldots\ldots+2n)=1$

(S2) : $\lim\limits_{n\to\infty}\frac{1}{n^{16}}\left(1^{15}+2^{15}+3^{15}+\ldots\ldots+n^{15}\right)=\frac{1}{16}$

(A) Only (S1) is true

(B) Both (S1) and (S2) are true

(C) Both (S1) and (S2) are false

(D) Only (S2) is true

Q311. JEE Main 2023 (Online) 1st February Morning Shift · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\left[\frac{1}{1+n}+\frac{1}{2+n}+\frac{1}{3+n}+\ldots+\frac{1}{2n}\right]$ is equal to

(A) 0

(B) $\log_e 2$

(C) $\log_e\left(\frac{2}{3}\right)$

(D) $\log_e\left(\frac{3}{2}\right)$

Q312. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\frac{3}{n}\left\{4+\left(2+\frac{1}{n}\right)^2+\left(2+\frac{2}{n}\right)^2+\ldots+\left(3-\frac{1}{n}\right)^2\right\}$ is equal to :

(A) 0

(B) $\frac{19}{3}$

(C) 19

(D) 12

Q313. JEE Main 2022 (Online) 26th July Morning Shift · MCQ (Single Correct Answer)

If $a=\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\frac{2n}{n^2+k^2}$ and $f(x)=\sqrt{\frac{1-\cos x}{1+\cos x}}$, $x\in(0,1)$, then :

(A) $2\sqrt{2}f\left(\frac{a}{2}\right)=f'\left(\frac{a}{2}\right)$

(B) $f\left(\frac{a}{2}\right)f'\left(\frac{a}{2}\right)=\sqrt{2}$

(C) $\sqrt{2}f\left(\frac{a}{2}\right)=f'\left(\frac{a}{2}\right)$

(D) $f\left(\frac{a}{2}\right)=\sqrt{2}f'\left(\frac{a}{2}\right)$

Q314. JEE Main 2022 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\frac{1}{2^n}\left(\frac{1}{\sqrt{1-\frac{1}{2^n}}}+\frac{1}{\sqrt{1-\frac{2}{2^n}}}+\frac{1}{\sqrt{1-\frac{3}{2^n}}}+\ldots+\frac{1}{\sqrt{1-\frac{2^n-1}{2^n}}}\right)$ is equal to

(A) $\frac{1}{2}$

(B) 1

(C) 2

(D) -2

Q315. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{r}{2r^2-7rn+6n^2}$ is equal to :

(A) $\log_e\left(\frac{\sqrt{3}}{2}\right)$

(B) $\log_e\left(\frac{3\sqrt{3}}{4}\right)$

(C) $\log_e\left(\frac{27}{4}\right)$

(D) $\log_e\left(\frac{4}{3}\right)$

Q316. JEE Main 2022 (Online) 24th June Evening Shift · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\left(\frac{n^2}{(n^2+1)(n+1)}+\frac{n^2}{(n^2+4)(n+2)}+\frac{n^2}{(n^2+9)(n+3)}+\ldots.+\frac{n^2}{(n^2+n^2)(n+n)}\right)$ is equal to :

(A) $\frac{\pi}{8}+\frac{1}{4}\log_e 2$ (B) $\frac{\pi}{4}+\frac{1}{8}\log_e 2$

(C) $\frac{\pi}{4}-\frac{1}{8}\log_e 2$ (D) $\frac{\pi}{8}+\log_e\sqrt{2}$

Q317. JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

If $U_n=\left(1+\frac{1}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)^2.....\left(1+\frac{n^2}{n^2}\right)^n$, then $\lim\limits_{n\to\infty}(U_n)^{\frac{-4}{n^2}}$ is equal to :

(A) $\frac{e^2}{16}$ (B) $\frac{4}{e}$ (C) $\frac{16}{e^2}$ (D) $\frac{4}{e^2}$

Q318. JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

The value of

$\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{r=0}^{2n-1}\frac{n^2}{n^2+4r^2}$ is :

(A) $\frac{1}{2}\tan^{-1}(2)$ (B) $\frac{1}{2}\tan^{-1}(4)$ (C) $\tan^{-1}(4)$ (D) $\frac{1}{4}\tan^{-1}(4)$

Q319. JEE Main 2021 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

The value of $\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{j=1}^{n}\frac{(2j-1)+8n}{(2j-1)+4n}$ is equal to :

(A) $5+\log_e\left(\frac{3}{2}\right)$ (B) $2-\log_e\left(\frac{2}{3}\right)$

(C) $3+2\log_e\left(\frac{2}{3}\right)$ (D) $1+2\log_e\left(\frac{3}{2}\right)$

Q320. JEE Main 2021 (Online) 25th February Evening Shift · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\left[\frac{1}{n}+\frac{n}{(n+1)^2}+\frac{n}{(n+2)^2}+.....+\frac{n}{(2n-1)^2}\right]$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) 1 (D) $\frac{1}{4}$

Q321. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\left(\frac{(n+1)^{1/3}}{n^{4/3}}+\frac{(n+2)^{1/3}}{n^{4/3}}+.....+\frac{(2n)^{1/3}}{n^{4/3}}\right)$
is equal to :

(A) $\frac{4}{3}(2)^{3/4}$ (B) $\frac{3}{4}(2)^{4/3}-\frac{3}{4}$

(C) $\frac{4}{3}(2)^{4/3}$ (D) $\frac{3}{4}(2)^{4/3}-\frac{4}{3}$

Q322. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\left(\frac{n}{n^2+1^2}+\frac{n}{n^2+2^2}+\frac{n}{n^2+3^2}+.....+\frac{1}{5n}\right)$ is equal to :

(A) $\tan^{-1}(2)$ (B) $\tan^{-1}(3)$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Q323. JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If $\lim\limits_{n\to\infty} \frac{1^a+2^a+.....+n^a}{(n+1)^{a-1}[(na+1)+(na+2)+.....+(na+n)]} = \frac{1}{60}$

for some positive real number a, then a is equal to :

(A) 7 (B) 8 (C) $\frac{15}{2}$ (D) $\frac{17}{2}$

Q324. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty} \left(\frac{(n+1)(n+2)...3n}{n^{2n}}\right)^{\frac{1}{n}}$ is equal to:

(A) $\frac{9}{e^2}$ (B) $3\log 3 - 2$ (C) $\frac{18}{e^4}$ (D) $\frac{27}{e^2}$

Q325. AIEEE 2005 · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty} \left[\frac{1}{n^2}\sec^2\frac{1}{n^2} + \frac{2}{n^2}\sec^2\frac{4}{n^2}.... + \frac{1}{n}\sec^2 1\right]$
equals

(A) $\frac{1}{2}\sec 1$ (B) $\frac{1}{2}$cosec 1 (C) tan 1 (D) $\frac{1}{2}$tan 1

Q326. AIEEE 2004 · MCQ (Single Correct Answer)

$Lim\limits_{n\to\infty} \sum\limits_{r=1}^{n} \frac{1}{n} e^{\frac{r}{n}}$ is

(A) $e+1$ (B) $e-1$ (C) $1-e$ (D) e

Q327. AIEEE 2003 · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty} \frac{1+2^4+3^4+....+n^4}{n^5}$ - $\lim\limits_{n\to\infty} \frac{1+2^3+3^3+....+n^3}{n^5}$

(A) $\frac{1}{5}$ (B) $\frac{1}{30}$ (C) zero (D) $\frac{1}{4}$

Q328. AIEEE 2002 · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty} \frac{1^p+2^p+3^p+.....+n^p}{n^{p+1}}$ is

(A) $\frac{1}{p+1}$ (B) $\frac{1}{1-p}$ (C) $\frac{1}{p} - \frac{1}{p-1}$ (D) $\frac{1}{p+2}$

Q329. JEE Main 2023 (Online) 6th April Evening Shift · Numerical

Let $f(x) = \frac{x}{(1+x^n)^{\frac{1}{n}}}, x \in \mathbb{R} - \{-1\}, n \in \mathbb{N}, n > 2.$

If $f^n(x) = (f \circ f \circ f ..... \text{ upto } n \text{ times})\,(x)$, then

$\lim\limits_{n\to\infty} \int\limits_0^1 x^{n-2}\,(f^n(x))dx$ is equal to _____.

Q330. JEE Main 2022 (Online) 25th July Morning Shift · **Numerical**

If $\lim\limits_{n\to\infty}\frac{(n+1)^{k-1}}{n^{k+1}}[(nk+1)+(nk+2)+\ldots+(nk+n)]$ $=33\cdot\lim\limits_{n\to\infty}\frac{1}{n^{k+1}}\cdot\left[1^k+2^k+3^k+\ldots+n^k\right]$, then the integral value of k is equal to

_____

Q331. JEE Main 2021 (Online) 16th March Morning Shift · **Numerical**

Let f : (0, 2) $\to$ R be defined as f(x) = $\log_2\left(1+\tan\left(\frac{\pi x}{4}\right)\right)$. Then, $\lim\limits_{n\to\infty}\frac{2}{n}\left(f\left(\frac{1}{n}\right)+f\left(\frac{2}{n}\right)+\ldots+f(1)\right)$ is equal to _____.

Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **D**	3. **B**	4. **B**	5. **B**
6. **B**	7. **B**	8. **C**	9. **C**	10. **D**
11. **C**	12. **B**	13. **C**	14. **B**	15. **B**
16. **A**	17. **D**	18. **D**	19. **B**	20. **B**
21. **C**	22. **D**	23. **B**	24. **D**	25. **A**
26. **A**	27. **A**	28. **C**	29. **C**	30. **A**
31. **B**	32. **B**	33. **B**	34. **C**	35. **B**
36. **D**	37. **C**	38. **A**	39. **B**	40. **B**
41. **B**	42. **D**	43. **A**	44. **C**	45. **B**
46. **D**	47. **A**	48. **B**	49. **B**	50. **A**
51. **A**	52. **A**	53. **C**	54. **C**	55. **C**
56. **C**	57. **A**	58. **C**	59. **C**	60. **C**
61. **B**	62. **B**	63. **B**	64. **C**	65. **D**
66. **B**	67. **B**	68. **B**	69. **C**	70. **C**
71. **B**	72. **B**	73. **B**	74. **A**	75. **C**
76. **A**	77. **B**	78. **D**	79. **A**	80. **A**
81. **A**	82. **C**	83. **A**	84. **A**	85. **A**
86. **D**	87. **B**	88. **D**	89. **A**	90. **B**
91. **D**	92. **A**	93. **A**	94. **D**	95. **B**
96. **B**	97. **C**	98. **C**	99. **D**	100. **B**
101. **A**	102. **D**	103. **C**	104. **C**	105. **D**
106. **B**	107. **C**	108. **C**	109. **B**	110. **B**
111. **A**	112. **D**	113. **D**	114. **C**	115. **C**
116. **B**	117. **C**	118. **D**	119. **B**	120. **D**
121. **A**	122. **C**	123. **B**	124. **C**	125. **B**
126. **C**	127. **B**	128. **C**	129. **B**	130. **C**
131. **A**	132. **A**	133. **A**	134. **A**	135. **D**
136. **B**	137. **B**	138. **C**	139. **D**	140. **B**
141. **A**	142. **B**	143. **A**	144. **C**	145. **C**
146. **A**	147. **D**	148. **B**	149. **A**	150. **D**

151. **D**	152. **A**	153. **D**	154. **C**	155. **C**
156. **A**	157. **B**	158. **C**	159. **B**	160. **A**
161. **D**	162. **A**	163. **C**	164. **C**	165. **A**
166. **C**	167. **C**	168. **A**	169. **C**	170. **A**
171. **B**	172. **A**	173. **A**	174. **D**	175. **D**
176. **D**	177. **C**	178. **D**	179. **B**	180. **C**
181. **B**	182. **D**	183. **C**	184. **A**	185. **D**
186. **C**	187. **C**	188. **A**	189. **D**	190. **A**
191. **C**	192. **A**	193. **B**	194. **C**	195. **C**
196. **A**	197. **A**	198. **A**	199. **B**	200. **C**
201. **C**	202. **A**	203. **B**	204. **D**	205. **D**
206. **A**	207. **C**	208. **C**	209. **D**	210. **B**
211. **D**	212. **C**	213. **B**	214. **D**	215. **B**
216. **B**	217. **B**	218. **D**	219. **C**	220. **B**
221. **D**	222. **C**	223. **C**	224. **D**	225. **D**
226. **D**	227. **A**	228. **B**	229. **D**	230. **B**
231. **D**				

Numerical

232. **12**	233. **192**	234. **9**	235. **210**	236. **2**
237. **64**	238. **16**	239. **36**	240. **2**	241. **9**
242. **17**	243. **8**	244. **12**	245. **64**	246. **112**
247. **32**	248. **23**	249. **65**	250. **1**	251. **48**
252. **8**	253. **219**	254. **8**	255. **15**	256. **176**
257. **27**	258. **5**	259. **155**	260. **12**	261. **6**
262. **2**	263. **41**	264. **18**	265. **575**	266. **32**
267. **6**	268. **14**	269. **13**	270. **63**	271. **1**
272. **12**	273. **20**	274. **27**	275. **22**	276. **2**
277. **104**	278. **10**	279. **385**	280. **12**	281. **24**
282. **8**	283. **5**	284. **16**	285. **3**	286. **21**
287. **6**	288. **6**	289. **1**	290. **34**	291. **5**
292. **4**	293. **5**	294. **1**	295. **16**	296. **8**
297. **512**	298. **1**	299. **1**	300. **16**	301. **406**
302. **1**	303. **2**	304. **19**	305. **3**	306. **21**

307. **1**	308. **1.50**

MCQ (More than One Correct Answer)

309. **C, B**

Out of syllabus

310. **B**	311. **B**	312. **C**	313. **C**	314. **C**
315. **B**	316. **A**	317. **A**	318. **B**	319. **D**
320. **A**	321. **B**	322. **A**	323. **A**	324. **D**
325. **D**	326. **B**	327. **A**	328. **A**	329. **0**
330. **5**	331. **1**			